

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

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VERSION **v. 1.1**

RELATED SUPPORT

- **TEMPLATE GUIDE Key Project Information & VPA Design Document v.1.1**

This document contains the following Sections

Key Project Information

Section A – Description of project

Section B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

Section D – Summary of Safeguarding Principles and Gender Sensitive Assessment

Section E – Summary of Local stakeholder consultation

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2 - Contact information of VPA Implementer (mandatory)

Appendix 3 - Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	GS11331
Title of Project	POA GS11330 Circle Gas LPG Smart Meter Program - Circle Gas LPG Smart Meter Program in Kenya-VPA01
Time of First Submission Date	01/09/2021
Date of Design Certification	28/07/2022
Version number of the VPA-DD	05
Completion date of version	16/08/2022
Coordinating/managing entity	ClimateCare Bidco Limited
VPA Implementer (s)	M-Gas Limited
Project Participants and any communities involved	Circle Gas Limited
Host Country (ies)	The Republic of Kenya
GS ID and Title of applicable Design Certified VPA	N/A
GS ID and Title of applicable Performance Certified VPA	N/A
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale ☒ Large Scale
Other Requirements applied	-GS4GG Principles and Requirements - Programme of Activity Requirements -Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 4.0 - CDM TOOL 30, version 3, Calculation of the fraction of non-renewable biomass

Methodology (ies) applied and version number	Reduced Emissions from Cooking and Heating : Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC) v4.0
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in section B.6)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Amount of GHGs emissions avoided or sequestered	1,949,899.00	tCO2e
8 Decent work and economic growth	Total number of employees earning above local minimum wage.	1,000	Number
7 Affordable and clean energy	Proportion of population with primary reliance on clean fuels and technology	385,623	Number
5 Gender equality	Average time saving associated with cooking time and fuel collection	3.2	hours/household/day
3 Good health and well being	Proportion of users that perceive a reduction in smoke levels.	100%	Percentage

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

The proposed project activity (Circle Gas LPG Smart Meter Program in Kenya) has been developed by Circle Gas Limited with the purpose of reducing the high demand for wood, charcoal and kerosene predominantly used for cooking in the baseline by end-users within the target areas. This will be achieved through distribution and installation of metered LPG cooking technology that is fitted with a smart meter to track and monitor LPG usage by the end-users. The project adopts a Pay as You Go (PAYG) approach to accelerate access to LPG by the end-users. The project activity also aims to contribute towards sustainable development goals. The specific description of the project activity is as follows:

- a) The project entails distribution and installation of the metered LPG cooking technology that is fitted with a smart meter to monitor LPG usage among end-users. The Pay as You Go approach will be adopted to increase access and continuous LPG usage among end-users
- b) The project developer is Circle Gas Limited, who will have the formal responsibility of the project activity.
- c) The VPA implementor for this project activity is M-Gas Limited and will be directly responsible for distribution of the project technology, maintaining the distribution database, maintenance of the units, training of sales and service team, collecting baseline information, and co-ordinating the annual monitoring process.
- d) The project activity is a voluntary action by the project developer.
- e) The goal of the project is to accelerate access to LPG which will significantly reduce GHG emissions, and other negative effects associated with use of traditionally unsustainable fuels such as wood, charcoal, and kerosene.

Goal and Objectives of the Project Activity

The goal of the project activity is to accelerate uptake of LPG for cooking which will significantly reduce GHG emissions, and other negative effects associated with use of traditionally unsustainable biomass fuels such as firewood and charcoal, and kerosene, normally used on inefficient cookstoves. The project technology entails use of meter LPG for cooking which will displace GHG emissions from the thermal energy consumption of households and commercial facilities.

Specific Objectives

1. To increase access to LPG for cooking within the project area which will replace traditional inefficient biomass (firewood and charcoal), and kerosene cookstoves used in the baseline.
2. To sensitize end-users on the benefits of using LPG fuel for cooking as compared to the highly polluting baseline fuels such as firewood, charcoal, and kerosene.
3. To improve community livelihoods through improved health and employment creation.

The location of the project activity

The location of the project activity is the Republic of Kenya. The geographical coordinates of Kenya are 0.0236° S, 37.9062° E. The target project areas are the 47 counties in Kenya.

The technologies/measures to be employed and/or implemented by the project activity

The project activity entails distribution and installation of the Circle Gas LPG Smart Meter technology. Circle Gas Limited has developed the smart meter technology to accelerate LPG uptake for cooking among end-users. The technology promoted entails installation of a complete metered LPG cooking technology, comprising of a single or double burner LPG cookstove, 13 kg LPG cylinder, a smart meter, and hose as a package to the end-user, and the end-user is required to pay for the gas. The smart meter allows for real-time monitoring of LPG consumption by the end-users.

The project activity also applies the Pay as You Go solution to ensure affordable, efficient, and convenient access to LPG by end-users. The PAYGO solution is an innovative approach that will enable end-users to adopt LPG for cooking by end-users, by allowing smaller purchases of fuel on an incremental, per use basis based on the available money without having to do a full cylinder refill. This will promote continuous use of LPG, since the barrier of having a full refill as required by conventional LPG systems will be removed.

By using the project technology and the PAYGO solution, the end-users will access clean energy and use energy efficient systems for cooking which will replace the use of inefficient baseline stove technologies.

The project boundary

The LPG smart meter technology is an investment of Circle Gas Limited and shall be implemented in the Republic of Kenya by the project implementer (M-Gas Limited). The project boundary includes the premises of the end-users (households and commercial facilities) and communities within the target areas of the project activity. The target area is the 47 counties in the Republic of Kenya. These counties include Mombasa, Kwale, Kilifi, Tana River, Lamu, Taita Taveta, Garissa, Wajir, Mandera, Marsabit, Isiolo, Meru, Tharaka-Nithi, Embu, Kitui, Machakos, Makueni, Nyandarua, Nyeri, Kirinyaga, Murang'a, Kiambu, Turkana, West Pokot, Samburu, Trans-Nzoia, Uasin Gishu, Elgeyo-Marakwet, Nandi, Baringo, Laikipia, Nakuru, Narok, Kajiado, Kericho, Bomet, Kakamega, Vihiga, Bungoma, Busia, Siaya, Kisumu, Homa Bay, Migori, Kisii, Nyamira and Nairobi. More details about the project boundary are provided in section B.3 of this VPA-DD.

Project duration

The project start date is 02/09/2020 which marks the date when the first metered LPG cooking technology was installed under the project. The crediting period start date is 02/09/2020. The length of the crediting period is 5 years (renewable twice).

Registration with other Carbon standards

The project activity is not registered with any other carbon standard, nor is it seeking registration under any other GHG programs, and hence there is no likelihood that double counting would occur. Additionally, each metered LPG cooking technology shall be assigned a unique customer identification number which gives a distinction on the end-users, while the smart meters are also serialised which will further ensure there is no double counting.

Carbon credits ownership

The legal ownership of the products generated belongs to Circle Gas Limited. The end-users' consent to the transfer of carbon credits generated from using the project technology at the time of installation of the Circle Gas metered LPG cooking technology. By signing the consent form, all carbon rights are transferred to Circle Gas Limited, who in return has signed a contract with the CME (ClimateCare Bidco Limited) to deliver VERs generated from the project activity.

Implementation plan

M-Gas Limited is the VPA implementor for the project activity in Kenya and will be directly responsible for maintaining the distribution database, collecting baseline information, and co-ordinating the annual monitoring process. M-Gas is also responsible for coordinating the distribution and installation of the units, maintenance services and management of the field sales and service teams.

The project activity's projection is to distribute 1,928,113 LPG units over a 5-year period, which will be installed within end-users (households and commercial) premises located within the project target areas.

The expected ex-ante annual average emission reductions are 1,949,899.00 tCO₂e, and the estimated total emission reductions for the first crediting period are 9,749,495.00.

Host Country regulations on emission reductions

The host country for the has not set any regulations setting a cap on the emission reduction generated by projects. There is also no possibility by the host country to trade emissions that include the scope of the proposed project.

Compliance with applicable Host Country's legal, environmental, ecological, and social regulations.

Table 2: Project's compliance with Host Country's legal, environmental, ecological, and social regulations

Applicable regulation	Provisions of the law	Project's compliance
2010 Kenya's Constitution	This is the principal guiding law in the country from which all the subsidiary laws are drawn from. It guarantees and gives every person in the country the right to enjoy a safe and clean environment.	The project activity seeks to accelerate access to clean energy which will benefit the end-users. Using the project technology will lead to improved health due to reduced indoor air pollution and will contribute towards reduced deforestation due to reduced firewood and charcoal usage. This is key in ensuring a clean and safe environment and human wellbeing.
Climate Change Act, 2016 ¹	The Climate Change Act (No. 11 of 2016) is the first comprehensive legal framework for climate change governance for Kenya. The objective of the Act is to "Enhance climate change resilience and low carbon development for	The project activity is promoting the use of efficient LPG technology. By using the technology, the pre-project practice of using the high carbon emitting non-renewable biomass and kerosene fuels by end-users for cooking are displaced. Through this, the project will contribute towards

¹ [No. 11 of 2016 \(kenyalaw.org\)](http://kenyalaw.org/No.11of2016)

	sustainable development of Kenya.”	mitigating climate change in Kenya.
Environmental Management and Coordination Act, 2009 ² (2015 amendment)	This is an Act of parliament to provide for the establishment of an appropriate legal and institutional framework for the management of the environment and for matters connected therewith and incidental thereto. Part VII on Environmental Audit and Monitoring section 58 specifically detail the need to undertake Environmental Impact Assessment of all projects likely to cause negative impacts to the environment as listed in the second schedule of the act. Further, part V of the Environmental Impact Assessment and therefore mandatory	According to the legal notice 31& 32 on EIA projects, LPG depots are categorized as medium risk project and EIA is required for such projects. Although the project is anticipated to have numerous positive environmental and social impacts, and EIA for the project’s depot was conducted in compliance with this act. The annual environmental audit for all depots has also been undertaken in compliance with the regulations and has been submitted as supporting evidence (See <i>Mukuru, Huruma and Kibera initial environmental audits 2021</i>).

² kenyalaw.org/kl/fileadmin/pdfdownloads/Acts/EnvironmentalManagementandCoordinationAct_No8of1999.pdf

	that an Environment Impact Assessment process.	
Environmental Management and Co-Ordination (E-Waste Management) Regulations, 2013 ³	These regulations apply to all categories of electrical and electronic equipment as set out in schedule 10 and facilities for transportation, recycling, re-use and recovery. Then regulations also set the guidelines for licensing of e-waste recyclers. Clause 11 requires that the generator shall ensure e-waste is segregated from other forms of waste and is taken to licensed refurbishers, collection centers or recyclers. Schedule 5 lists products which shall be taken into account for the purpose of the regulation, among which batteries are included.	The project activity is promoting use of the smart meter LPG technology. The meters make use of lithium-ion-based rechargeable batteries, which have a lifespan of up to 25 years. After the lifespan, the batteries will need to be safely disposed. The VPA implementer shall be responsible for ensuring all batteries that reach the end of their lifespan are collected and taken to an approved collection center for safe disposal as required under the regulations.

³ [Draft E-waste Regulations-1.pdf \(nema.go.ke\)](#)

Employment Act, 2007 ⁴ .	The Act declares and defines the fundamental rights of employees, to provide basic conditions of employment of employees, to regulate employment of children, and to provide for matters connected with the foregoing. The act provides that an employer shall promote equal opportunity in employment and strive to eliminate discrimination in any employment policy or practice. Further, an employer shall pay his employees equal remuneration for work of equal value.	The project shall ensure compliance to this act by providing equal employment opportunities to all people without discrimination. Further, the project shall monitor the number of people employed and their remuneration to ensure equal pay for equal value of work done is paid, and ensure all employees are paid within the country's national minimum wage ⁵ .
Standards Act, 2012 ⁶	The purpose of the act is to promote the standardization of the specification of commodities, and to provide for the	The project activity has complied with the provisions of the act by undertaking the required tests on the smart meter and have obtained the Euro Cert

⁴ [Microsoft Word - Paged Employment Act Cap. 226 - No. 11 of 2007 .doc \(kenyalaw.org\)](#)

⁵ [Kenya Minimum Wage - World Minimum Wage Rates 2022 \(minimum-wage.org\)](#)

⁶ [Microsoft Word - Paged Standards Act Cap. 496 - No. 17 of 1973 .doc \(kenyalaw.org\)](#)

	standardization of commodities and codes of practice; and to establish a Kenya Bureau of Standards.	standardization mark from the European Inspection and Certification Company. The project technology has been found to comply with the Essential health and safety requirements relating to the design and construction of products intended for use in potentially explosive atmospheres (<i>see the Circle Gas Euro Cert certificate has been provided as supporting evidence⁷</i>). The smart meter has also obtained a calibration certification from the Kenya Bureau of Standards (KEBS)⁸ which is the National standard body in the host country. The certificate confirms that the smart meter has achieved the required accuracy test (<i>See M-Gas KEBS certificate provided as supporting evidence</i>).
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⁷ The certificate is under Kopa Technology who was the initial owner of the technology.

⁸ [Kenya Bureau Of Standards \(KEBS\)](#)

Regulatory framework for LPG supply

Table 3: Summary of LPG Supply regulatory framework in Kenya

Applicable regulation	Provisions of the law	Project's compliance
The Energy Act, 2019 ⁹	The act consolidates the laws relating to energy, provides for National and County Government functions in relation to energy, provides for the establishment, powers, and functions of the energy sector entities; promotion of renewable energy; exploration, recovery, and commercial utilization of geothermal energy; regulation of midstream and downstream petroleum and coal activities; regulation, production, supply and use of electricity and other energy forms. The Act established the Energy and Petroleum Regulatory Authority	The project activity entails distribution and installation of metered LPG cooking technology fitted with a smart meter to be used for cooking by end-users and is therefore regulated by this act. To ensure compliance with the law, the project implementor has obtained the License for Import, Export and Wholesale of LPG in Bulk from the Energy and Petroleum Regulatory Authority (EPRA) and this has been provided as a support document (<i>see M-Gas Import export & wholesale of LPG in bulk document</i>). The project implementor has also contracted an authorized company (Rift Gas) for the storage and refill of LPG in compliance

⁹ [EnergyAct No.1of2019.PDF \(kenyalaw.org\)](#)

	(EPRA) whose part of its mandate is to regulate the importation, refining, exportation, transportation, storage, and sale of petroleum and petroleum products except for crude oil.	with the act. The company has obtained the required license from EPRA which has been provided as a support document (<i>see Rift Gas Storage & filling of LPG in cylinders (RG)</i>).
The Petroleum (Liquefied Petroleum Gas) Regulations 2019 ¹⁰	The act provides that any person who wishes to undertake liquefied petroleum gas business shall make an application for a license to the Authority or its licensing agents in the prescribed manner. The act also provides that a person shall not fill liquefied petroleum gas into cylinders unless he is the brand owner or has prior written consent from the brand owner. Further, A person licensed to undertake	The project implementor has ensured compliance with this act by obtaining the License for Import, Export and Wholesale of LPG in Bulk from EPRA which has been provided as a support document. All the gas cylinders used by the project belong to M-Gas and have been branded 'M-Gas'. The project has contracted an authorized company which is in charge of conducting inspection and refill of the cylinders in compliance with the act.

¹⁰ [The Petroleum \(Liquefied Petroleum Gas\) Regulations 2019 - Energy and Petroleum Regulatory Authority \(epra.go.ke\)](https://epra.go.ke/The-Petroleum-(Liquefied-Petroleum-Gas)-Regulations-2019)

	liquefied petroleum gas business shall investigate any accident occurring with respect to a licensed activity and shall within fourteen days or any such extended period as approved by the Authority from the date of the accident submit a report containing the — (a) cause of the accident. (b) effects of the accident; and (c) proposed remedial measures and timelines thereof.	The project implementor has hired and trained fire marshals and will work in close partnership with the local authorities to handle any event of fire within its areas of operation. The project developer has established a fire policy to guide its fire preparedness (<i>see the provided M-Gas Fire Safety Policy</i>). The depots have also conducted a fire inspection and clearance certificate issued by the Nairobi City County (<i>See M-Gas Fire Certificate</i>). The project has also invested in safety enhancement of the project technology to minimize fire risks and shall comply with the provisions of this act.
The Occupational Safety and Health Act, 2007 ¹¹	Key purpose of this act is to provide for the safety, health and welfare of workers and all persons lawfully present at workplaces.	The project implementor (M-Gas) shall ensure compliance with the provisions of this act. A fire inspection shall be conducted annually by the

¹¹ [Microsoft Word - THE OCCUPATIONAL SAFETY AND HEALTH BILL vellum rev.doc](#)

		County Government, fire department. The depot has also been registered as workplaces with the relevant authority, the Directorate of Occupational Health and Safety (DOSHS). The workplace registration certificate has been provided in compliance with the act (<i>see M-Gas workplace registration renewal</i>), which has been provided as a support document.
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Corrective Action Requests (CARs) and Forward Action Requests (FARs)

1 CAR and 7 FARs were raised during the Preliminary review stage, and these have been addressed as summarized below:

Likely CAR # 1: The start date shall be justified in line with the definitions of the start date. For distributed technology projects only, the start date is defined as the implementation of the first unit under the project. The relevant evidence for the start date shall be provided at the time of validation. The VVB shall provide opinion on the same at the time of validation.

The PP has confirmed the project start date as 02/09/2020. This is the date when the first unit of the project technology was distributed. The PP has submitted the end-user contracts signed for the units installed as proof of project start date (*see Circle Gas-Proof of project start date document*)

FAR #1: The PD shall supply supporting data for all parameters in time for validation/design review, or allocation may be delayed. This includes and is not limited to: ER spreadsheets, individual study calculations, survey results, study reports etc.

The project developer conducted baseline scenario survey and Kitchen performance test and this data has been provided (*See Baseline data and KPT data spreadsheets*). The ex-ante emission reduction calculations have also been done in line with the methodology as provided under the *GS11331 Ex-ante ERs spreadsheet*. Other supporting documents for the parameters have also been submitted for review.

FAR # 2: The new version (version 4.0) of the applied methodology (TPDDTEC) is now published on the GS website. The PD shall note that there is no grace period for the last version (version 03) and the new project shall use the latest version 04 of the applied methodology. As the preliminary review already started with the older version, hence the following requirement applies.

"GHG EMISSIONS REDUCTION & SEQUESTRATION PRODUCT REQUIREMENTS § 8.2.2"
The methodology and tool version applied at the time of first submission (preliminary review) may be used for Design Certification as long as the Project is submitted for validation within six months after the time of first submission for Preliminary Review. If this condition is not met, the latest version of the methodology and applicable tool(s) available at the time of submission for validation shall be applied". Hence the above requirements shall be taken care of during the validation.

The methodology applied by the project activity has been upgraded to TPDDTEC (Version 4.0) as required. This has been discussed on the VPA-DD and applied for the Ex-ante calculations. The project activity is eligible under the latest methodology and the applicable emission reduction calculation method is method 3.

FAR # 3: As more than 5 SDG Impacts are listed, hence the VVB shall assess the same in detail at the time of validation.

The project has multiple benefits and will contribute to several SDG goals. However, due to the costs and efforts required to monitor SDG outcomes, the PD has revised the listed SDGs and the project activity shall now seek contribution towards SDGs 3,5,7,8 and 13. The methodological choices/approaches for estimating the SDG Impact is detailed in section B.6.1 of the VPA-DD and the monitoring plan in section B.7.

FAR # 4: Stakeholder Consultation: The VVB shall check and confirm that the inputs from women and marginalised groups were undertaken as part of the Consultation.

The PP can confirm that input was sought from women and marginalised groups during the stakeholder consultations. As evident in the invitation tracking sheet, both men and women were invited to the LSC meeting. Targeted invitations were made to marginalised groups including women and youth groups, persons living with disabilities, environmental and community groups and civil societies representing the vulnerable in the community.

As indicated in the LSC participants list, the meeting was attended by 20 men and 15 women. The attendees included representatives of marginalized groups including women, youth, and community-based organisations.

FAR # 5: As the project is retroactive, hence the physical meeting shall be incorporated into the SFR. The VVB shall confirm the same at the time of validation.

The PP can confirm that a physical meeting for the project activity was held on 05/08/2021 in line with the Gold Standard Stakeholder Engagement guidelines. The purpose of the meeting was to discuss the PoA design and the implementation of the project. A stakeholder feedback round was also conducted between 24/11/2021 to 24/01/2022 to seek further input from the stakeholders who were invited to the meeting. Details of the SFR are included in section E.1 of the VPA-DD and the stakeholder consultation report.

FAR # 6: The VVB shall determine on the requirement of opinion of an expert stakeholder for the following at the time of validation:

Principle 4.1 - Sites of Cultural and Historical Heritage

Principle 4.2 - Forced Eviction and Displacement

Principle 4.3 - Land Tenure and Other Rights

Principle 4.4 - Indigenous Peoples

Principle 8.1 - Impact on Natural Water Patterns/Flows

Principle 8.2 - Erosion and/or Water Body Instability

Principle 9.10 - High Conservation Value Areas and Critical Habitats

Principle 9.11 - Endangered Species

The PP has conducted a detailed safeguarding principles assessment for the project activity which was also discussed with the stakeholders during the meeting, as presented in Appendix 1. The project activity entails distribution and installation of metered LPG cooking technology to end-users for cooking. From the assessment done, the project activity has not been assessed to have any significant negative implication on the listed principles, and hence opinion of an expert is not required. An environmental and social impact assessment was done for the project. All anticipated impacts were assessed and required mitigations put in place for impacts deemed as significant. A license was issued by the approving authority as proof that the project has demonstrated sufficient procedure for mitigating impacts, and the same will be monitored through the annual environmental audits as required.

For the principles where the project is anticipated to have any potential impact as indicated in section D.1 of the VPA-DD, the project developer shall put in place measures to mitigate the impacts, and this will be monitored and reported in the monitoring report.

FAR # 7: The PD shall claim Emission Reductions only related to end-use energy efficiency improvements. The Emission Reductions related to the difference in carbon content in the fuels between baseline scenario and project scenario shall not be claimed. Hence the PD shall meet the following requirements at the time of validation.

Paragraph 9.1.1 (Section 9) of the "Community Services Activity Requirements, version 1.2" that stipulates "The emission reductions related to the difference in carbon content between a non-renewable fuel and a less carbon-intensive non-renewable fuel used for substitution shall NOT be eligible".

As the project replaces biomass (fuelwood and charcoal), which is more carbon intensive, hence the PD shall clarify as how the project may be considered as eligible under the GS4GG.

Applicability Criteria No. 2.2.1 (f) that stipulates that "Project activities making use of solid fossil fuel in the project scenario or other improved fossil fuel cookstoves meeting certain conditions described in the footnote to Table 1 (e.g., switch from three-stone fire biomass stoves to LPG stoves) may only claim emission reductions for

energy efficiency improvement aspect and shall assume the same baseline and project fuel for emission reduction calculations".

Hence the VVB shall ensure that the above requirements are met at the time of validation.

The PP can confirm that since the project technology is promoting the use of fossil fuel for cooking (LPG), claims will only be made to Emission Reductions related to end-use energy efficiency improvements in compliance with the methodology requirement for fossil fuel-based projects. The PP has demonstrated the eligibility of the project activity under the latest methodology (see table 7 included in the VPA-DD), the GS4GG Principles and Requirements, and the Community Services Activity Requirements. As provided under the applied methodology (TPDDTEC v4.0), fossil fuels projects are required to estimate emission reduction calculations based on method 3 which allows claim only on energy efficiency improvement. The PP has applied this method in the calculation of the Ex-ante emission reductions as required and has only claimed end-use energy efficiency (see the GS11331 Ex-ante ER spreadsheet).

A.1.1. Eligibility of the project under the PoA

Table 4: Eligibility of the VPA under the PoA eligibility criteria

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
1.	<i>Geographical boundaries of VPAs consistent with the geographical boundary of the PoA;</i>	The VPA shall be located within the PoA boundary	Location and boundary shall be specified in the specific VPA-DD stating that the location is limited to an area within the host country of the PoA. Documentation: The geographical boundary of the VPA has been identified as the Republic of Kenya. The VPA is within the geographical boundary of the host

			country included in the PoA.
2.	<i>Conditions to avoid double counting of Impacts, such as unique identifications of product and end user locations (e.g., programme logo)</i>	Identification of each metered LPG cooking technology through the smart meter serial number and unique customer identification number. Each VPA will show that it is exclusive to the PoA and not registered as another project activity or VPA under another PoA.	Each metered LPG cooking technology shall have a unique smart meter serial number and a customer identification number ensuring that they are uniquely identifiable to a particular VPA. An end user agreement with each participating end-user shall also be signed at the time of installation. Documentation: End user agreements
3.	<i>Conditions to confirm that VPAs are neither registered as project activities with other offset Schemes, included in other registered PoAs, nor the project activities that have been deregistered</i>	Confirmation that the VPA is not registered with other carbon standards	A declaration by the Project developer has been provided to confirm that the VPA is neither registered as a project activity with GS or any other standard or as a VPA of another PoA. The appropriate registries (Gold Standard and CDM) have been cross-checked to demonstrate this (see <i>Circle Gas registration with other program document</i>).
4.	<i>Specification of the technology/measure such as the level and type of service, as well as performance specification based on, inter alia, testing/certification</i>	The VPA shall involve the distribution and installation of metered LPG cooking technology to end-users currently cooking with non-renewable biomass and kerosene.	The distribution database for each VPA will be cross-checked to ensure that only the specified project technology is included in the VPA. The target beneficiaries shall be household and

			commercial end-users. Documentation: Distribution database and product sheet.
5.	<i>Conditions to check the crediting period start dates of V/CPAs through documentary evidence</i>	The VPA crediting period start date shall not be earlier than the PoA crediting period start date.	The VPA crediting period start date has been given as 02/09/2020 (project start date). The VPA crediting period start date is the same as PoA crediting period start date (crediting period start date for the first VPA)
6.	<i>Conditions to ensure compliance with the applicability of the applied methodologies, the applied standardized baselines and the other applied methodological regulatory documents</i>	The VPA shall meet the applicability requirements of the TPDDTEC v 4.0 methodology and GS4GG Principles and Requirements. Baseline surveys shall be conducted to determine the baseline scenario for each VPA.	The applicability of the methodology is justified in Section B.3 and applies to each VPA. Baseline surveys were conducted to confirm the baseline scenario. The detailed discussion on the baseline scenario is provided under section B.4 of the VPA-DD. Documentation: Baseline survey data
7.	<i>Conditions to ensure that V/CPAs meet the requirements for demonstration of additionality</i>	Each VPA within the PoA shall demonstrate additionality through the Gold Standard additionality criteria or the latest version of the CDM Methodological Tool for the demonstration and assessment of additionality.	Assessment of additionality has been done at VPA level as outlined in section B.5 of the VPA-DD. Documentation: VPA-DD
8.	<i>Conditions to ensure no diversion of official development assistance</i>	The project developer shall confirm that in case of public funding there	A declaration of non-use of ODA has been completed and

		shall not be diversion of Official Development Assistance.	submitted for the VPA. Documentation: Signed ODA form.
9.	<i>Target group (e.g. domestic/commercial/industrial, rural/urban, grid connected/offgrid), and where applicable, distribution mechanisms (e.g. direct installation)</i>	The target groups for the VPAs shall include domestic and commercial end-users.	The project activity shall distribute the metered LPG cooking technology to both domestic and commercial end-users. Documentation: Distribution records
10.	<i>Conditions related to sampling requirements for the PoA</i>	Sampling will be done during: 1. The baseline surveys 2. During the annual monitoring process. As stipulated in the methodology, the project will comply with the set sampling requirements as follows: 1) Baseline surveys The baseline survey requires in person interviews with a robust sample of end-users without project technologies that are representative of end users targeted in the project activity. Representative and random sampling shall be carried out following these guidelines for minimum sample size: • Group size <300: Minimum sample size 30 or population size, whichever is smaller.	The VPA shall apply the Guidelines for sampling and surveys stipulated in the methodology. The project implementor conducted 130 baseline surveys where in person interviews were conducted with end-users without the project technologies. Documentation: • Baseline survey sampling sheet and baseline survey data spreadsheet with all the responses from the baseline survey.

		• Group size 300 to 1000: Minimum sample size 10% of group size. • Group size > 1000 Minimum sample size 100. 2) Annual monitoring: the field performance tests shall satisfy the 90/10 rule, where the confidence interval is 90% and a 10% margin of error.	• Monitoring survey spreadsheet with all the monitored data and calculations from each VPA.
11.	<i>Conditions to confirm that technologies in V/CPAs are eligible (refer to A.3 above)</i>	All VPAs included in the PoA shall meet the methodological requirements, Community Service Activity requirements and the general GS4GG eligibility requirements specified in Section B.3 of the PoA-DD. Eligible VPAs shall promote and distribute the technology specified in this PoA-DD which typically aim at accelerating LPG uptake through installation of metered LPG cooking technology and will use the Pay as You Go approach.	The VPA has specified the technologies being promoted in the VPA-DD and this meets the set eligibility requirement. Documentation: VPA-DD
12.	<i>Conditions to be met by each VPA regarding SDG outcomes assessment</i>	Each VPA shall conduct an SDG Outcomes assessment and shall positively contribute to at least 3 of the sustainable development goals identified in the PoA-DD.	The SDG outcomes assessment has been done at VPA level. The identified SDGs will be monitored and reported in the monitoring report.

			Documentation: - SDG outcomes assessment in the VPA-DD. - Monitoring report
13.	<i>Conditions to be met by each VPA regarding safeguarding principles</i>	Each VPA shall conduct a safeguarding principles assessment in line with the Gold Standard Safeguarding principal assessment requirements. The project activity shall employ safeguards that aim at protecting the social, economic, and environmental aspects of the people and areas it operates.	The CME has conducted a safeguarding principles assessment and can confirm that the VPA has adhered to the safeguarding principles. Any deviation or incident that affects the project safeguards shall be monitored and reported in the monitoring report. Documentation: Safeguarding principles assessment in the VPA-DD - Safeguarding principles monitored in the monitoring report.
14.	<i>Stakeholder consultation and engagement</i>	Stakeholder consultation and engagement will be done at the VPA level. The PoA is multi country which makes it impossible to do PoA level stakeholder consultations.	A physical stakeholder consultation process for the VPA was conducted on 05/08/2021 with the key stakeholders in line with GS4GG Stakeholder consultation and engagement guidelines. A second round of consultation the stakeholder feedback round was conducted from 24/11/2021 to 24/01/2022 to seek further input from the stakeholders. Documentation: Stakeholder consultation report

15.	<i>Legal Ownership</i>	The VERs generated by the project activities shall be the sole ownership of the CME. During the installation of the units, a Carbon Waiver Form will be signed by the purchaser stating that the rights to the carbon credits have been relinquished to the project developer.	The rights to the carbon credits generated by the project activity shall be owned by Circle Gas Limited. End-users shall consent to the transfer of the ownership of the carbon rights by signing an end-user agreement provided at the point of installation. Circle Gas has also signed an Emission Reductions Purchase Agreement with the CME (ClimateCare) on the transfer of VERs generated by the project. Documentation: The end user agreements relinquishing carbon credits generated to the project developer.
16.	<i>Project eligibility under the methodology</i>	Demonstrate if project is pre identified as eligible by being referenced in Gold Standard Activity Requirements, Impact Quantification Methodologies or Product Requirements	The proposed project activity is eligible under the methodology "<i>Reduced Emissions from Cooking and Heating: Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC) v4.0</i>". The project technology falls within the scope of "<i>Improved solid fossil fuel cookstoves or other improved fossil fuel cookstoves meeting certain conditions, where only emission reductions from efficiency improvement are</i>

			<i>eligible</i>". The Applicable emission reduction calculation is Method 3. The project activity is also eligible under section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2): Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section. The project entails on the ground installation of the LPG units among end-users and is therefore eligible under this requirement. As per section 4.1.3 of <i>GS4GG Principles & Requirements</i> (Version 1.2), "A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it, or it's referenced in the Gold Standard Product Requirements". The project type is automatically eligible for Gold Standard Certification. Section 3 of <i>GS4GG Community Services</i>
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			<i>Activity Requirements (Version 1.2): the project activity falls under section 3.1.1 (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end user of the products and services are clearly identified and when the physical intervention is required at the user end."</i> <i>Section 5.1.1 of GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0): part (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end user of the products and services are clearly identified and when the physical intervention is required at the user end."</i>
17.	<i>Emission reduction cap</i>	Demonstrate the activity is not located in a host country, region, locality, or state that has an	The project activity shall be implemented in the Republic of Kenya and the country does not

		emission reduction cap enforced OR has the possibility to trade emissions that include the scope of the proposed project	have an emission reduction cap, nor does it trade emission that include the scope of the proposed project activity. This is detailed in section A.1 of the VPA-DD.
18.	<i>Host Country's legal requirement</i>	Demonstrate that the project is in compliance with applicable Host Country's legal, environmental, ecological and social regulations	The project activity meets the host country's legal, environmental, ecological, and social regulations. This is demonstrated in detail in section A.1 of the VPA-DD.

Eligibility with GS4GG Requirements

The proposed project falls under section 3.1.1 of *GS4GG Principles & Requirements* (Version 1.2), section 3 of *GS4GG Community Services Activity Requirements* (Version 1.2) and section 5 of *GS4GG GHG Emissions Reduction & Sequestration Product Requirements* (Version 2.0) with the following eligibility criteria:

Table 5: Eligibility of the Project with GS4GG requirements

Eligibility Criteria	Justification
(a) Types of Projects Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section. Section 3.1.1 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Pre-identified CSA project types are a) Renewable energy; b) End-use energy efficiency; c) Waste management and handling; d) Water, sanitation, and hygiene (WASH).	The project activity involves distribution and installation of metered LPG cooking technology among end-users. This is aimed at accelerating access to metered LPG cooking technology to reduce use of baseline biomass fuels such as firewood and charcoal, and kerosene. Therefore, the project activity falls under section 3.1.1 (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end user of the products and services are clearly identified and when the physical intervention is required at the user end." Besides, the project activity falls under section 5.1.1 (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end user of the products and

Section 5.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> The Following Project types are eligible for issuance of Gold Standard VERs: a) Renewable Energy Supply; b) End-Use Energy Efficiency Improvement; c) Waste Handling & Disposal; d) Land Use and Forests.	services are clearly identified and when the physical intervention is required at the user end” of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i>. As per section 4.1.3 of <i>GS4GG Principles & Requirements (Version 1.2)</i>, “A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it, or it’s referenced in the Gold Standard Product Requirements”. The project type is “End-Use Energy Efficiency Improvement” and the Impact Quantification Methodology to be applied is “Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC) v4.0”. Section 2.2.1 (f) provides that “Project activities making use of solid fossil fuel in the project scenario or other improved fossil fuel cookstoves meeting certain conditions described in the footnote to Table 1 (e.g., switch from three-stone fire biomass stoves to LPG stoves) may only claim emission reductions for energy efficiency improvement aspect and shall assume the same baseline and project fuel for emission reduction calculations”. The project activity shall only claim emission reduction based on
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	this requirement and is hence eligible for Gold Standard Certification.
(b) Location of Project: Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Projects may be located in any part of the world. Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Gold Standard VER Projects may be located in any host country or state. However, where host countries or states have mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either: (a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occurs or (b) has committed to retiring eligible units equal to the quantity of Gold	The project activity shall be located in the Republic of Kenya (refer to section A.2 of the VPA-DD for detailed project area and boundary). There are no mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in Kenya, which based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) is an eligible host country.

Standard VERs. Refer to Annex A of this document.	
(c) Project Area, Project Boundary and Scale: Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. To avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects).	The project activity location is the Republic of Kenya (refer to section A.2 of the VPA-DD for detailed project area and boundary) and there are no mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in Kenya, which based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) is an eligible host country. The project activity entails installation of complete metered LPG cooking technology among end-users within the project boundary. The project technology entails use of smart meters to monitor real time use of LPG used for cooking by the end-users. The Pay as You Go approach is applied to allow end-users to purchase fuel equivalent to available money. This is a unique technology and solution that is being applied in the project boundary by the project developer and no other Gold Standard or other voluntary or compliance standard programme of a similar nature operates in the project boundary. Based on this, no overlapping is expected and hence no

Section 3.1.2 of <i>GS4GG Community Services Activity Requirements (Version 1.2)</i> Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. The definition of scale is the same for all Projects, except Microscale. Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> Gold Standard VER Projects may be located in any host country or state. However, where host countries or states have mandatory operational schemes to reduce GHG emissions in any form (e.g. cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either: (a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occurs or (b) has committed to retiring eligible units equal to the quantity of Gold Standard VERs. Refer to Annex A of this document. Section 9.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i>	potential for double counting of impacts. The project activity boundary is defined based on the methodology "<i>Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC)</i> (Version 4.0)" and is limited to the Republic of Kenya (refer to section B.3 of the VPA-DD). The project activity location is the Republic of Kenya (refer to section A.2 of the VPA-DD for detailed project area and boundary). As indicated above, the host country does not have an emission reduction cap, nor does it trade emission that include the scope of the proposed project activity. The project activity is defined as large scale since the expected aggregate energy savings 8,643.46 GWh which
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GSVER Projects may be registered as 'large scale', 'small scale' (for the applicability of methodologies and tools only) or 'microscale'. Scale is defined in the relevant Activity Requirements or where these do not exist then per following paragraphs. Section 9.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> All Projects exceeding the small-scale thresholds are defined as large scale. Small scale projects are defined in accordance with CDM project standard for project activities, as below; (a) Type 1: Renewable energy Projects: maximum output capacity of 15 MW(e) or 45MW(th). (b) Type 2: End-use energy efficiency project improvement: activities that reduce energy consumption, on the supply and/or demand side, with a maximum energy saving of 60 GWh per year (or an appropriate equivalent) in any year of the crediting period. In this context, for project activities that improve thermal energy efficiency, the maximum energy saving of 60 GWh(e) per year is equivalent to 180 GWh(th) per year saving; (c) Type 3: Other project activities: project involves technologies such Safe Water Supply, Waste management, etc.	exceeds 60 GWh per year per year in fuel input.
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not included in Type I or Type II that result in GHG emission reductions not exceeding 60,000-ton CO₂e per year in any year of the crediting period.	
(d) Host Country Requirements Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations. Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Gold Standard VER Projects may be located in any host country or state. However, where host countries or states have mandatory operational schemes to reduce GHG emissions in any form (e.g. cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either:	The project activity location is the Republic of Kenya (refer to section A.2 of the VPA-DD for detailed project area and boundary). The host country has not set any mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.). Based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Kenya is an eligible host country. A detailed assessment on the project's compliance with the applicable Host Country's legal, environmental, ecological, and social regulations is summarized in tables 2 and 3. The implementation of the project activity is fully in line with the relevant energy policies of Kenya. The project technology enables end-users to access the metered LPG technology for cooking. The implementation of this project activity will accelerate uptake of LPG for cooking which is in line with Kenya's vision on sustainable energy resources. Besides, Kenya submitted its updated Nationally Determined Contribution

(a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occurs or (b) has committed to retiring eligible units equal to the quantity of Gold Standard VERs. Refer to Annex A of this document.	(NDC) on 24/12/2020, which commits to abate GHG emissions by 32% by 2030 relative to the business as usual (BAU) scenario of 143 million tCO₂e¹². The project is in compliance with the applicable Host Country's legal, environmental, ecological and social regulations, benefits the local residents and will help in contributing towards Kenya's National emission reductions target.
(e) Contact Details Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organization (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its	The Project Participants' contact details are provided in Appendix 2 of the VPA-DD. Contact information of the project participants of this VPA-DD. The project developer (Circle Gas) and the VPA implementor (M-Gas) are legally registered to operate in the respective host countries as proof that they are in good standing. The companies' certificates of incorporation are provided as supporting evidence (<i>see CG and M-Gas Certificate of incorporation</i>).

¹²

[https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Kenya%20First/Kenya's%20First%20%20NDC%20\(updated%20version\).pdf](https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Kenya%20First/Kenya's%20First%20%20NDC%20(updated%20version).pdf)

Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	
(f) Legal Ownership Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising. Section 3.1.4 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the	Each metered LPG cooking technology end-user, who is the default owner of emission reductions, is notified on the need to waive ownership of the emission reductions generated from using the project technology upon installation of the units. The waiver will be signed with each end-user when installing the metered LPG cooking technology to make sure the project developer reserves the legal ownership of the emission reductions generated by the project activity. This is done by signing/consenting to the end-user agreement which contains the details regarding the transfer of the carbon rights. Refer to section A.1.2 of the VPA-DD for further details. Moreover, the ownership of the products generated from the project activity was discussed during the stakeholder consultation meeting. The stakeholders discussed the need for transfer of ownership of the carbon rights, and it was determined that the project owner had the undisputed legal ownership of the carbon credits generated from use of the project technology.

ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	
(g) Other Rights Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas.	The project activity involves installation of complete metered LPG cooking technology to end-users within the project target area which is aimed at accelerating access to metered LPG cooking technology. The complete LPG unit comprises of a 13kg cylinder, the smart meter, LPG cookstove, smart card, and hose. The gas refill shall be done by a contracted authorized company hence the project activity is not putting up an LPG plant (<i>see provided Storage and Refilling of LPG cylinders license issued by the regulatory authority to the contracted firm</i>). The project activity does not involve any activity that causes alteration of any resource, or contested legal rights and other disputes, therefore the need for acquiring any specific legal right is not applicable.

h) Official Development Assistance (ODA) Declaration Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification. Section 6.1.1 and 6.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Projects are ineligible for carbon crediting under Gold Standard if the ODA assistance is provided to the project under the condition that the credits generated by the Project will be transferred, either directly or indirectly, to the donor country providing ODA support.	The host country (Kenya) is named in the OECD ODA recipient list¹³. The project developer has signed the ODA declaration and confirms that no ODA has been provided for the project activity (see provided <i>Circle Gas_ODA Declaration document</i>)
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¹³ [DAC List of ODA Recipients for reporting 2021 flows June 2021.xlsx \(oecd.org\)](#)

Project Developer submitting a Project located in a country named by the OECD Development Assistance Committee's ODA recipient list shall sign and submit the ODA Declaration.	
i) Eligible Greenhouse Gases Section 4.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version2.0)</i> Only Carbon Dioxide (CO₂), Methane (CH₄) and/or Nitrous Oxide (N₂O) are eligible for Gold Standard crediting, provided Projects comply with Gold Standard Requirements and eligibility criteria.	The project activity only considers the reduction of Carbon Dioxide (CO₂), Methane (CH₄) and Nitrous Oxide (N₂O) emissions for Gold Standard crediting. Refer to section B.3 of the VPA-DD for further details.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

The legal ownership of the products generated by the project activity belongs to the Project developer (Circle Gas Limited). The terms and conditions document for using the project technology outlines the procedure and need for transfer of the carbon rights. This information is provided to the end-users during installation of the unit and registration to the project. During the registration process, each customer is assigned a unique customer identification number. During installation of the project technology, each customer with a smartphone is required to use their unique customer number to create their user account and then consent to the terms and conditions, which includes the carbon waiver.

Since the project activity applies the IoT Cloud Gateway which facilitates communication between the end-users and the project developer, a detailed record of all the end-users who have consented to the transfer of carbon rights can be downloaded and provided as evidence that the carbon rights transfer is in place. This approach is deemed appropriate and effective to contain the volume of paperwork since the projected number of installations over the course of the crediting period is quite high.

For the end-users without smartphones, the terms and conditions will be provided in hardcopy for consent and manual signing during installation of the units. By signing the waiver, the rights to the carbon credits generated by the project are transferred to Circle Gas Limited and they remain uncontested. The signed hardcopy forms will then be scanned and stored by the VPA implementor. Circle Gas has also signed an Emission Reductions Purchase Agreement with the CME (ClimateCare) on the transfer of VERs generated by the project. The Terms and Condition template has been provided as a support document (*see provided Circle Gas End-user contract confidential document*).

A.2 Location of project

The project boundary for the VPA is the Republic of Kenya.

The geographical coordinates of Kenya are 0.0236° S, 37.9062° E.

The target areas are the 47 counties of the Kenya which are represented in the map below.

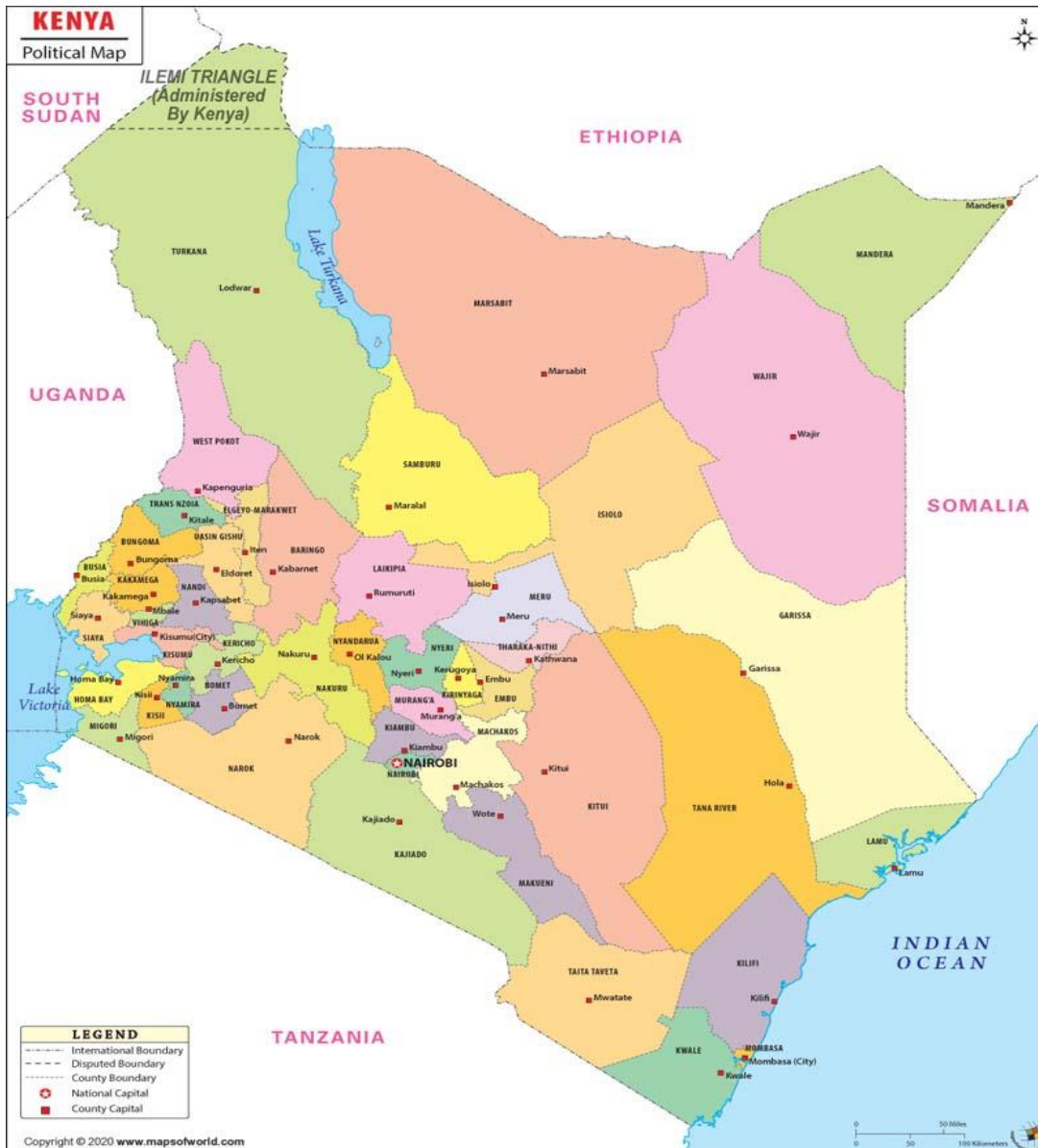


Figure 1: The Administrative map of Kenya

Source: [Kenya Africa Map \(mapsofworld.com\)](http://Kenya Africa Map (mapsofworld.com))

A.3 Technologies and/or measures

The project activity entails distribution and installation of the Circle Gas LPG Smart Meter Pay as You Go technology. Circle Gas Limited has developed the smart meter technology which is fitted on LPG cylinder to monitor LPG use by end-users. The project also adopts the Pay as You Go solution to accelerate LPG uptake for cooking by end-users in the targeted project area.

M-Gas Limited (the VPA implementor) shall provide the LPG units applying any of the three package models as follows:

- i. Package A: provide a 2-burner stove, smart card, hose, 13 kg LPG cylinder and the smart meter.
- ii. Package B: provide 1 burner stove (for end-users without a space for 2 burner stove), smart card, hose, 13kg LPG cylinder and the smart meter.

In the first phase, the project developer has been providing the complete unit to the end-users at zero cost and were only required to pay for their gas. However, in the second phase the project developer will be selling the cookstoves at a reasonable price to the end-user which is expected to create ownership by the customer to take care of the gas stove. The other components of the unit will be delivered at zero cost.

The project developer has provided a flexible stove repayment solution where:

- (i) the end-user can make a one-off payment for the 2 burner cookstove, or
- (ii) Pay for the stove in 24 months instalment with a weekly payment of Ksh.25 (\$0.25)

The Smart Meter is a proprietary technology of Circle Gas Limited and releases gas to customers until the funds balance in their embedded digital wallet runs out. The Smart meter is topped up using an electronic payment solution which reports gas consumption back to Circle Gas systems to support billing, monitoring and scheduling of replacement of depleting cylinders. The project developer has partnered with

Safaricom¹⁴ (the largest mobile and internet service provider in Kenya) in implementation of the project activity. The relationship enables joint technology development, expertise sharing, co-creation of solutions, and market support between the two firms. The smart meter connects to Safaricom's Narrow Band Internet of Things (NB-IoT) network and accepts payment through mobile money service (M-Pesa¹⁵). M-Pesa Kenya is an electronic mobile money service that allows you to store, send and receive money on your mobile phone. With an M-Pesa account, one can transfer cash and shop for products and services. One can also withdraw cash by visiting an agent (typically in a corner shop) or transfer it to others from your phone. This makes it a quick, safe, and simple way to make payments and handle money. Once an end-user credits the M-Gas account using M-Pesa, the amount reflects on the meter and gas is released allowing the end-users to enjoy convenient cooking.



Figure 2: The Circle Gas Smart Meter that is fitted to LPG cylinders.

The IoT Cloud Gateway facilitates communication with smart meters, while the customer app keeps customers informed of their gas consumption and helps them in

¹⁴ [Mobile Phones, Tablets, Mobile Money Solutions & Telecommunication Services - Safaricom](#)

¹⁵ [M-PESA \(safaricom.co.ke\)](#)

managing their cost of cooking. The mobile money service (M-Pesa) provided by Safaricom¹⁶ allows customers to pay for gas through a simple transaction on their mobile phone.

Inergy is a backend database, account management and customer relationship management solution that monitors the consumption of gas and schedules replacement of gas cylinders before the customers run out. The Inergy app assigns tasks to the field service teams and tracks workloads. The field sales team uses the Smart Sale app which helps them to map prospective customers, especially in areas with inadequate data and location mapping.

Below is as summary of the applications, intended user and their functionality.

Table 6: Summary of the Narrow Band Internet of Things (NB-IoT) applications employed by the project technology

Application for	Used by	Functionality
Identifying new customers	Sales Agents	✓ Register customer leads; ✓ Schedule follow-up calls; ✓ Register new customers; and ✓ Also used by the management team to assign tasks to sales agents based on their current location/ site.
Verification and Installation	Installation Agents and Technicians	✓ Conduct home visit to check the customer home is compliant to the safety requirements for LPG; ✓ Customer training; and ✓ Install cylinder, smart meter and cook stove.
Customer Data App	Customer	✓ Monitor gas usage and balance; ✓ Purchase gas credit with mobile money; and ✓ Training.
Stock Management	Operations team	✓ Track every cylinder from purchase to delivery; and ✓ Schedule cylinder refilling.

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A summary of how the metered LPG cooking technology installation works is as follows.

1. M-Gas installs a new gas cylinder, smart meter, and stove in the end-users' premises. M-Gas ensures the system is properly installed and conducts a safety brief.
2. The end-user pays for the gas you need using M-PESA.
3. The smart meter helps release the gas and the end-user enjoys cooking using LPG.
4. When the cylinder is nearly empty the smart meter sends an alert to M-Gas.
5. M-Gas schedules a free delivery.
6. M-Gas staff visits the end-user and replaces the empty cylinder.

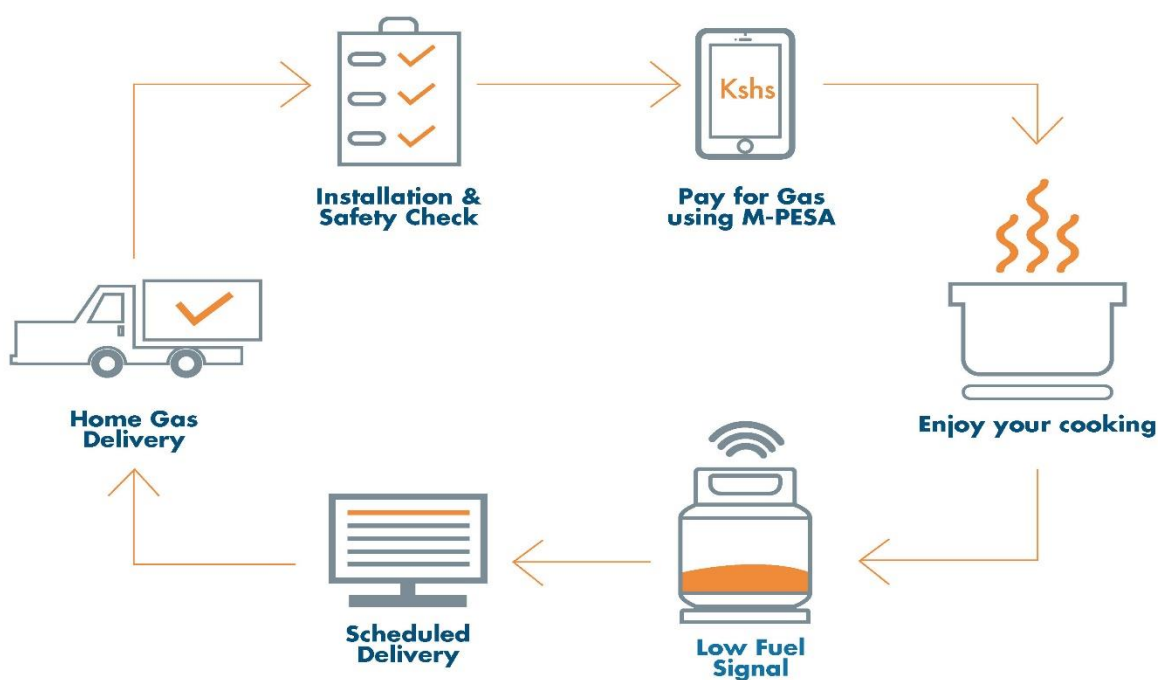


Figure 3: A schematic illustration of how the M-Gas Pay as You Go system works.

Safety measures employed by the project

The project activity has put in place measures to enhance the safety of the end-users using the project technology as compared to conventional LPG systems in the following ways:

- **The smart card:** Each end-user is provided with a smart card which is required to activate gas usage. An end-user cannot light the gas without swiping the smart card hence minimising the risk of children using the gas in absence of their parents or guardians.
- **Smart meter automatic switching off:** the smart meter is automatic and once the end-user activates the smart meter by swiping the card and fails to light the stove, the smart meter automatically switches off after 15 seconds since it detects gas flow, and no lighting.
- **Leakage detection:** the smart meter can detect an abnormal flow of gas in case of a leakage and automatically shuts down. This reduces the risk of fire incidences associated with gas leakage.
- **Focus on 13kg gas:** the project only focuses on installation of 13kg cylinder which have a release valve, that prevents it from bursting in case of a fire. The cylinders shall be revalidated and inspected regularly to ensure they are in good condition and safe to use.
- **24-hour customer service:** M-Gas customer service centres are open throughout, and all the end-users have been provided with the toll-free customer service numbers to contact M-Gas in case of any emergency. An end-user can call to report in case of a gas leak or fire, or any other needed support and will get assistance immediately.
- **Training:** M-Gas has employed trained and certified fire marshals whose responsibility is to respond to any issues of fire within the community in the targeted areas.
- **Installation procedure:** The installation of the units is done by a trained team to ensure appropriate procedure is followed. During installation, the LPG unit is installed opposite the door, below the window, away from the bedroom/sleeping areas and away from a corner that has no lighting. All this is done to ensure that the installation is done in a well-ventilated place and away from places where things can easily catch fire like the bedroom.

The technical specifications of the metered LPG cooking technology is as described below.

Smart meter specifications

Parameters	Smart meter
Manufacturer name	KopaTech
Ambient temperature	+10°C~+45°C
Working temperature	84kPa~106.7kPa
Maximum working pressure	750kPa in the cylinder
Outlet pipe size	11.5mm
Outlet pressure	3kPa
Maximum gas flow	1.5kg/h
Operational Lifetime	7 years
Working conditions	Should not be exposed directly to the sunlight or water
Fuel type	LPG
Warranty	1 year

Source: Circle Gas smart meter manual

Stove specifications

Parameters	
Manufacturer name	Different based on brand ¹⁷
Stove type	Double or single burner stoves
Gas consumption (kh/h)	Different based on brand but approximated at 0.5 kg/h (double), 0.27kg/h (single)
Dimension (H*W*D) mm	Different based on brand but approximated 120*720*390 (double), 140*300*390 (single)
Weight (kgs)	Different based on brand but approximated at 7kg (double burner), and 4kg (single burner)
Gas inlet	Inner diameter approximated at 9.5 mm
Operational Lifetime	7 years
Fuel type	LPG

¹⁷ The project developer is making use of different brands of LPG cookstoves in the market hence the manufacturers will vary. Description provided is based on the generic stove model utilized by the project.

Efficiency	50-60% ¹⁸
Warranty	6 months

Source: Circle Gas stove user manual

Sustainable development benefits

The aim of the project activity is to replace traditional biomass (fuelwood and charcoal), and kerosene cookstoves used in the baseline for cooking, with metered LPG cookstoves which will deliver impacts across the following multiple areas as follows:

- **Improved health:** Using LPG for cooking displaces the use of highly emitting fuels such as firewood, charcoal and kerosene which disproportionately impacts on women and children due to smoke and other indoor air pollutants causing respiratory illnesses. Indoor Air Pollution (IAP) is one of the largest health risk factors for mortality in Kenya with about 21,560 deaths attributed to IAP annually¹⁹. Clean burning liquefied petroleum gas (LPG), although a fossil fuel, emits low levels of Black Carbon and minimal PM2.5 concentrations, typically meeting WHO Indoor Air Quality Guideline levels for health²⁰ (SDG 3).
- **Environment & Climate Change:** using traditional inefficient biomass and kerosene stoves for cooking consumes high amount of fuel and emits GHG emissions to the atmosphere. Using the project technology for cooking is expected to reduce CO₂ and black carbon emissions hence reducing the impacts of climate change. Additionally, a shift towards use of LPG cookstoves will displace inefficient baseline cooking technologies and practices hence reducing deforestation within targeted project areas (SDG 13).
- **Gender equality and empowerment of women:** Cooking on traditional biomass inefficient stoves (firewood and charcoal) consumes a lot of fuel leading to constant need to source the fuel. Consequently, fuel collection is usually done by women and girls, and this requires more time and effort which

¹⁸ Studies show LPG stoves have an efficiency of 50-60%. Since the project is not promoting improved fossil fuel stoves, and is using different brands of stoves, no tests have been done and the conservative value of 50% will be applied for reporting purposes. ([A2A Kenya CleanCookingStudy July2021.pdf \(ambitiontoaction.net\)](#))

¹⁹ [MoE-2019-Kenya-Cooking-Sector-Study-compressed.pdf \(eedadvisory.com\)](#)

²⁰ [World Health Organization Guidelines for Indoor Air Quality \(foobot.io\)](#)

takes away their productive hours. The project technology promotes use of metered LPG cooking technology which is more efficient and will save on the end-users' time since there will be no need for collecting fuel and cooking time will equally be shortened. This will lead to improved use of time at home especially for women who can venture in other income generating activities and give children more time for study (SDG 4 & 5).

- **Money and Time savings:** Families spend a significant portion of their time and income on fuel collection and purchase. This is largely due to use of inefficient technology and cooking practices which waste a lot of fuel, leading to more demand. As fuel becomes scarce, expenditure on fuel becomes more significant due to the increase in fuel prices and this leads to more income being spent on meeting the end-user energy provision. The project technology promotes use of metered LPG cooking technology, which will require less fuel compared to the baseline technologies. This will provide meaningful money and time savings for the end-users (SDG 1).
- **Affordable and Clean Energy:** Affordable energy for cooking is an essential form of energy access for end-users which will be achieved through implementation of this project activity. A significant proportion of end-users will benefit from the LPG technology promoted by the project thus increasing the number of people using clean and affordable energy within the targeted project area (SDG 7).
- **Job creation:** the project will create direct employment opportunities to many people especially the youths who will be employed as field sales and service teams, and the overall management of the project. Indirect opportunities will also arise through the LPG production and distribution channel and is expected to benefit many people (SDG 8).

A.4 Scale of the project

As outlined in section 9.1.1 and 9.1.2 of GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0), the project is a large-scale project. The annual average thermal energy savings is expected to be 8,643.46 GWh which is more than 180 GWh (see GS11131 Ex-ante ERs tab Energy savings).

A.5 Funding sources of project

The project activity does not receive, nor will it receive any public funding. To affirm the same, the project developer has signed the ODA declaration form which has been submitted to the Gold Standard as a supporting document.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1 Reference of approved methodology (ies)

The applied methodology by the VPA is “Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC), Version 4.0”

This is the latest versions of the methodology which have been applied at the time of project listing.

Other tools and guidelines applied include:

- GHG Emissions Reduction & Sequestration Product Requirements, Version 2.1
- Community Services Activity Requirements, Version 1.2
- Programme of Activity Requirements, Version 1.2
- Stakeholder Consultation and Engagement Requirements, Version 1.2
- Stakeholder Consultation and Engagement Guidelines, Version 1.2
- GS4GG Principles & Requirements, Version 1.2
- CDM Tool 30, Calculation of The Fraction of Non-Renewable Biomass, Version 3.0
- Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 4.0
- SDG Impact Tool v.1.0

B.2 Applicability of methodology (ies)

The VPA involves the distribution and installation of metered LPG cooking technology and adopts the use of the Pay as You Go LPG solution. The smart meter devices are a timely modern technology that will increase access to LPG uptake in Kenya and help the end-users to achieve improved efficiency and fuel savings while providing an improved level of cooking service.

The project activity applies the GS methodology “*Technologies and Practices to Displace Decentralized Thermal Energy Consumption, Version 4.0*”.

This project qualifies under the applied methodology which provides the following:

Table 7: Eligibility of the project under the applicable methodology (TPDDTEC v.4.0)

Eligibility Criteria	Justification for the Project Activity
1. Section 2.1.1. provides that this methodology is applicable to project activities that introduce technologies and/or practices that reduce or displace greenhouse gas (GHG) emissions from the thermal energy consumption of households and/or residential, institutional, industrial, or commercial facilities. Throughout the methodology the term 'technology' should be read as the single or multiple technologies and/or practices applied in the project activity.	The project activity entails distribution of metered LPG cooking technology, which comprise of 13kg LPG cylinder, single or double burner cookstove, smart meter and smart card, and hose. The smart meter tracks real time fuel consumption among end-users. Use of this technology will displace GHG emissions from the thermal energy consumption of households and commercial facilities. As required under the methodology, although the project is promoting fuel switch from baseline fuels (firewood, charcoal, and kerosene), the project shall only claim emission reductions from efficiency improvement. The project is therefore eligible and emission reductions shall be calculated using method 3.
2. Section 2.1.2: Where there is no installation of improved devices and project claims emission reductions from improved practices only, project shall provide a detailed discussion of the chosen monitoring approach to demonstrate that quantified emission reductions result exclusively from the practices introduced by the project activity.	The project activity entails installation of metered LPG cooking technology which is more efficient compared to the baseline stoves technologies and emissions reductions shall be claimed from improved efficiency. In the host country, affordability remains a major barrier to LPG access for cooking among end-users. While LPG costs are frequently lower relative to purchasing biomass fuels, an unreliable supply of

	LPG and the financial outlay required to pay for the upfront costs is prohibitive for many. Further, among those that can afford the initial cost of the complete LPG unit, insecure incomes and precarious financial circumstances prevent some end-users from seeking full cylinder LPG refills, leading to its unsustainable usage as a primary cooking fuel. As a result, most LPG end-users abandon its use and revert to use of the baseline fuels. The Project is providing an innovative solution of addressing these barriers by providing the complete metered LPG cooking technology comprising of single or double burner LPG cookstove, 13 kg LPG cylinder fitted with the smart meter technology, hose, and a smart card as a package to the end-user at zero cost during the first phase of the project. In the second phase the cookstoves will be sold at a subsidised cost and end-users can pay the full amount upfront or through 24 months instalment. The end-user is required to pay for their gas using the Pay as You Go solution to allow them to purchase fuel equivalent to available money.
3.Section 2.1.3: Project may involve progressive distribution of technology where implementation of the technology may occur in a gradual manner and adoption can increase over the project's	The project activity is projected to progressively distribute 1,928,113 LPG units among end-users within the project boundary over a 5-year period. The implementation of the project requires

crediting period.	significant logistical and finance planning in setting up the depots, producing the smart meters, procuring cookstoves and managing the field sales and service teams to ensure effective operation. Due to this, the technology will be distributed progressively until all targeted areas are reached. The distributed units shall be recorded in the project's distribution database.
4. Section 2.2.1 (a) Project shall choose a technology design that has predictable performance in that it is proven to be efficient and durable under field conditions; for cookstoves, the rated thermal efficiency shall be at least 20%	The smart meter technology adopted by the project is a modern, efficient, and durable technology. The smart meter technology has received the Eurocert SA certification and has been certified to comply with essential health and safety requirements relating to the design and construction of products intended for use in potentially explosive atmospheres. The certificate has been provided as supporting evidence. The technology has also been tested by the Kenya Bureau of Standards (KEBs) and have been certified to comply with the required calibration standards. Since the project developer is not producing or distributing improved LPG cookstoves, the LPG stoves to be used by the project will be the conventional brands available in the market. The project developer will be responsible for getting the stoves from the manufacturers and will be responsible for ensuring the stoves are durable, of good

	quality and compatible with the other components of the project device. The cookstoves are expected to have a rated thermal efficiency of 50-60% which is more than the required minimum efficiency of 20%.
5. Section 2.2.1 (b) The technology shall have continuous useful energy output of less than 150kW per unit, where “continuous useful energy output” is defined as per Parameter ICS 2.	The project involves distribution and installation of LPG units, specially designed with a smart meter technology to track real time fuel consumption during cooking. The LPG cookstoves adopted by the project are the conventional brands in the market and these have been found to have on average a continuous useful energy output 3.6kW (single burner) and 5.65Kw (double-burner) which is less than 150kW per unit (<i>refer to Mika user manual</i>). As required under the methodology, although the project is promoting fuel switch from baseline fuels (firewood, charcoal, and kerosene), the project shall only claim emission reductions from efficiency improvement.
6. Section 2.2.1(c) The project activity is implemented by a project developer and can include additional project participants listed in Appendix 2 of the VPA-DD template. The individual households and institutions may be represented collectively by community organizations,	The project activity is being implemented by Circle Gas Limited (project developer) while ClimateCare Bidco Limited is the project representative. The project developer shall work with project implementors on the ground implementation of the VPA. The VPA implementer for this VPA is M-Gas

etc., but do not individually act as project participants.	Limited as detailed under Appendix 2 of the VPA-DD. End-users benefiting from the project shall not be individually included as project participants.
7.Section 2.2.1(d) The project developer must design incentive mechanism(s), which should be effective as fast as possible, for the elimination of inefficient baseline stoves that are replaced by the project cooking devices and describe the incentive mechanism(s) in the PDD/VPA-DD at the time of validation.	The project activity has designed several incentives to encourage uptake and continuous and sustained use of the project technology, for the eliminated use of inefficient baseline stoves as follows: 1. The project implementor shall provide the complete metered LPG cooking technology (13kg LPG cylinder, single or double burner cookstove, smart meter and smart card, and hose) to the end-user at no cost or at a subsidized cost for the cookstove. The end-users would be required to pay for their gas. This will eliminate the barrier associated with the high initial capital required to obtain a complete LPG system in the host country hence accelerate uptake of the project technology. 2. The project activity is adopting the use of Pay as You Go solution to ensure affordable, efficient, and convenient access to LPG by end-users. The Pay as You Go (PAYG) solution, which is an innovative approach, will enable households to buy fuel through smaller purchases on an incremental, per use basis. This will ensure that the end users can purchase gas based on the available

	money without having to do a full refill, which is one of the factors contributing to resumption in use of inefficient baseline stoves. PAYG supports continued LPG use even through periods of economic downturn. 3. The project shall adopt use of the smart meter technology to monitor fuel consumption during cooking. Study shows that households using LPG in Kenya travel up to 5.3 km on average to purchase the fuel than kerosene users (2.9 km). The long distance contributes to continued use of inefficient baseline fuels. Using the smart meter enables monitoring of fuel usage and the refill is scheduled before gas completely runs out, hence end users don't have to travel to purchase fuel. 4. The project implementer has established a robust field sales and service team. This will ensure efficient delivery of the LPG cylinders and maintenance of the meters/cookstoves. The LPG delivery will be done at no extra cost which further makes the project attractive for end-users.
8. Section 2.2.1 (e): To avoid double counting or double claiming, the project developer must: i. clearly communicates its ownership rights and intention of claiming the emission reductions resulting from the project activity to the following parties	The project technology has been developed by Circle Gas Limited. The project technology under the project activity will be distributed by Circle Gas Limited, through its project implementer (M-Gas) and no other external parties have been contracted to distribute or

by contract or clear written assertions in the transaction paperwork: • all other project participants. • project technology manufacturers; and • retailers of the project technology or the renewable fuel in use; and ii. inform and notify the end users that they cannot claim emission reductions from the project, and	retail this technology. Therefore, there is no likely risk of double counting. The legal ownership of the carbon credits generated from use of the project technology belongs to Circle Gas Limited. This information is provided to the end-users during installation of the unit and registration to the project through the contract signed with the end-users. During the registration process, each customer shall be assigned a unique customer identification number. Each customer with a smartphone is required to use their customer number to create their account on the smart app and then consent to the terms and conditions, which includes the carbon waiver agreement. Since the project activity applies the IoT Cloud Gateway which facilitates communication between the end-users and the project developer through the smart meters, the database will be downloaded from the app with a list of all the end-users who have consented to the transfer of carbon rights. This will include their name, customer number, telephone number, date and time and shall be provided as evidence. For the end-users without smartphones, the terms and conditions will be provided in hardcopy for consent and signing during installation of the units. By
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iii. exclude from the project activity, cooking devices included in any other voluntary market or CDM project activity/PoA and strive not to displace the cooking devices of another CDM or voluntary project/PoA.	signing the waiver, the rights to the carbon credits generated by the project are transferred to Circle Gas Limited and they remain uncontested. The End-user contract as a support document. The project activity is neither registered with any other voluntary market or CDM project activity/PoA, nor does it strive to displace the cooking devices of another CDM or voluntary project/PoA. A confirmation letter has been signed from the project developer reaffirming the same (<i>see provided Circle Gas-Registration with another program letter</i>).
9.Section 2.2.1 (f) Project activities making use of solid fossil fuel in the project scenario or other improved fossil fuel cookstoves meeting certain conditions described in the footnote to Table 1 (e.g. switch from three-stone fire biomass stoves to LPG stoves) may only claim emission reductions for energy efficiency improvement aspect and shall assume the same baseline and project fuel for emission reduction calculations.	The project activity entails distribution and installation of metered LPG cooking technology, which allows real time fuel consumption monitoring among end-users. The units shall make use of liquified petroleum gas (LPG) and no solid fossil fuel will be utilized by the project. As required under the methodology, although the project is promoting fuel switch from baseline fuels (firewood, charcoal, and kerosene), the project shall only claim emission reductions from energy efficiency improvement and shall assume the same baseline and project fuel for emission reduction calculations. This will be calculated using method 3 as required in the methodology.
10.Section 2.2.1 (h): Adequate evidence	Using the project technology is expected

is supplied to demonstrate that indoor air pollution (IAP) levels are not worsened compared to the baseline, and greenhouse gases emitted by the project fuel/stove combination are estimated with adequate precision. Furthermore, for projects where cooking will move from outdoor to indoor or where the project technology reduces ventilation (For example, changing from a stove with chimney to improved stove with no chimney), indoor air pollution (IAP) levels shall not worsen in the project compared to the baseline, including PM 2.5 and carbon monoxide (CO) emissions. This may be demonstrated before project Design Certification or during project operation using the certification resulting from of a manufacturer's test, report of field testing of the technology's PM 2.5 and carbon monoxide (CO) emissions, report of lab testing of the technology, or results of modelling of the technology's operation under field conditions. If none of these are available, reference from published literature or report by independent agencies may be used as evidence, provided it is not more than 5 years old.	to significantly improve the indoor air pollution level compared with the baseline scenario. The project seeks to accelerate LPG uptake which will promote a shift from using highly emitting baseline fuels such as firewood, charcoal, and kerosene. The WHO recognizes that shifting from solid fuels to cleaner energy technologies – for instance, liquid petroleum gas (LPG), biogas or solar power generation – can potentially yield the largest reduction in indoor air pollution levels while minimizing environmental impacts of energy production and consumption in general²¹. The PD has provided a description of IAP level of the project technology and has demonstrated that Indoor air pollution (IAP) levels shall not be worsened in the project scenario compared to the baseline as detailed on ICS 4.
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²¹ [WHO | Indoor air pollution and household fuels](#); [Household air pollution and health \(who.int\)](#)

11. Section 2.3.1: The project shall not undermine or conflict with any national, sub-national or local regulations or guidance for thermal energy supply or fuel supply or use. The project shall document the national, regional and local regulatory framework for provision of thermal energy services of the type of the project provided in the project boundary (parameter ICS 7).	The Project developer has undertaken an assessment of the national, sub-national or local regulations or guidance for thermal energy supply or fuel supply or use and has demonstrated that the project activity has not undermined any regulation as detailed in Tables 2 and 3.
12. Section 2.3.2: If the expected technical life of project technology (parameter ICS 3) is shorter than the crediting period, the project developer shall describe measures to ensure that end users are provided replacement technology of comparable quality at the end of the technical life, by either replacing with comparable or better technology, or retrofitting essential parts with performance guarantee. If neither of the prior conditions can be demonstrated, no emission reductions can be claimed for the technology after its technical life has ended.	The project technology is expected to have a lifespan of 7 years, with proper use and maintenance. The LPG cylinders shall be inspected by the firm contracted to do the refilling of the cylinders before refill to ensure they are in good condition. The project technology will be checked regularly by the VPA implementer to ensure proper functioning.

B.3 Project boundary

The project boundary is specified in Section 3.2 of the applied GS methodology.

The aim of the project activity is to introduce use of electric induction stoves for cooking which will promote displacement of non-renewable fuels including firewood, charcoal, and fossil fuels traditionally used for cooking by households in the project boundary.

The project boundary for the project activity is therefore defined as the end-user premises where the eCook units will be installed and the environs within which the

project technology is utilized. The target area for this project activity is defined as the Kinddom of Cambodiablic of Kenya, identified as the project boundary.

In reference to requirement ii on the project boundary, although the project activity promotes use of liquefied petroleum gas which is a processed fuel, the scope of the project activity does not entail putting up an LPG processing and refill plant. This service has been subcontracted to an external company which is authorized by the regulatory authority to do processing and refill of LPG cylinders.

The greenhouse gases included in or excluded from the project boundary are shown in Table below.

Table 8: Greenhouse Gases included in the project boundary.

Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Combustion of non-renewable biomass (fuelwood and charcoal) and Kerosene for cooking in inefficient cookstoves. (Cooking with inefficient baseline stoves).	CO ₂	Yes	Major source of emissions
		CH ₄	No	Minor emission source excluded as a conservative measure
		N ₂ O	No	Minor emission source excluded as a conservative measure
Project scenario	Combustion of fossil fuel (LPG) for cooking in more efficient cook stoves. (Cooking with fuel-efficient stoves).	CO ₂	Yes	Major source of emissions
		CH ₄	No	Minor emission source excluded as a conservative measure
		N ₂ O	No	Minor emission source excluded as a conservative measure

B.4 Establishment and description of baseline scenario

According to Section 3.4 and 3.5 of the methodology applied, TPDDTEC, version 4.0, the baseline scenario is defined as the existing baseline technology/practice use and fuel consumption patterns for the type of service provided by the project technology in the population targeted for adopting the new project technology, i.e., "target population".

The selection and description of the baseline scenario has been informed by the findings of the Baseline Scenario Survey conducted for the project activity. The baseline scenario survey was conducted with 130 households who were randomly selected from the target areas, and who were not using the project technology.

Out of the 130 surveys conducted, 62 households (47.7%) were using firewood for cooking as the primary fuel, followed by kerosene and charcoal at 26.9% and 25.4% respectively as shown in table 9.

Table 9: Primary baseline fuel analysis

Primary Baseline fuels	outcome	%
Firewood	62	47.7%
Charcoal	33	25.4%
Kerosene	35	26.9%
		100.0%

Source: Baseline survey data spreadsheet tab baseline fuel consumption

The survey also assessed the secondary fuels used by the households and from the findings, the predominant secondary fuel was charcoal at 54.5% followed by kerosene at 27.3% as shown in table 10.

Table 10: Secondary baseline fuel analysis

Secondary Baseline fuels	Outcome	Percentage
Charcoal	6	54.5%
Kerosene	3	27.3%
LPG	1	9.1%
Other	1	9.1%
Total	11	100.0%

Source: Baseline survey data spreadsheet tab baseline fuel consumption

From the baseline survey finding, the baseline scenario for the project activity has been identified as the use of non-renewable biomass fuels (firewood and charcoal), and kerosene to meet thermal energy requirements for household cooking. The surveys considered distinct baseline scenarios for the targeted end-user populations to factor in both urban and rural end-users who were found to consume significantly different fuels

The predominant baseline stove technologies were identified to be traditional inefficient stoves or a conventional device without a grate or a chimney i.e., with no improved combustion air supply or flue gas ventilation. From the survey finding, 42/80 households (32.3%) were using traditional 3 stone firewood stove as the primary baseline stove, followed by kerosene and non-improved charcoal stove at 26.9% and 20.8% respectively as summarized on table 11.

Table 11: Primary baseline stove analysis

Primary Baseline stoves	Outcome	Percentage
Traditional 3 stone firewood stove	42	32.3%
Improved firewood stove	20	15.4%
Non improved charcoal stove	27	20.8%
Improved charcoal stove	6	4.6%
Kerosene stove	35	26.9%
		100.0%

Source: Baseline survey data spreadsheet tab baseline fuel consumption

Pictorial representation of sample baseline stove technologies



Figure 4: A traditional 3 stone firewood stove at Ben Salaash Household in Narok County, Kenya

Source: Baseline surveys 2021



Figure 5: A traditional 3 stone firewood stove at Jane Jerita Household in Siaya County, Kenya

Source: Baseline surveys 2021



Figure 6: A non-improved charcoal stove at Judith Adhiambo Household in Migori County, Kenya
Source: Baseline surveys 2021



Figure 7: A kerosene stove at Hawo Adan Household in Nairobi County, Kenya
Source: Baseline surveys 2021

For the secondary stove users, 63.6% were using Non improved charcoal stoves followed by kerosene stoves at 27.3% as summarized in table 12.

Table 12 Secondary baseline stove analysis

Secondary baseline stoves	Outcome	Percentage
Traditional 3 stone firewood stove	0	0.0%
Improved firewood stove	0	0.0%
Non improved charcoal stove	7	63.6%
Improved charcoal stove	0	0.0%
Kerosene stove	3	27.3%
LPG stove	1	9.1%

Source: Baseline survey data spreadsheet tab baseline fuel consumption

The proposed project activity will distribute LPG units which are specially designed with the smart meter technology to monitor real time fuel usage. This will accelerate uptake of LPG for cooking and promote a switch from use of the baseline fuels including non-renewable biomass (firewood and charcoal) and kerosene and replace use of traditional inefficient baseline stoves technologies currently used for cooking in the baseline. This will consequently lead to a reduction in the emissions associated with use of non-renewable biomass and kerosene.

However, as required under method 3 of the applicable methodology, where the project introduces fossil fuels, only emission reductions from efficiency improvement are eligible, i.e., the baseline and project fuel emission factor must be assumed to be the same.

a) Baseline scenario survey

The baseline scenario survey provides critical information on target population characteristics, baseline technology use, fuel consumption, leakage, and sustainable development indicators.

The Project developer undertook a baseline scenario survey to collect the required baseline data from the target population, baseline technology use and fuel consumption, and the sustainable development indicators.

The survey was undertaken between 01/11/2021 to 03/11/2021, with 130 households who had been randomly selected. The surveys were undertaken by a group of 27 trained enumerators derived from Community Health Volunteers (CHVs) from the respective county health departments where baseline surveys were done (See *provided baseline survey participants list*).

Prior to undertaking the survey, a protocol was developed outlining a plan as to how the survey and kitchen performance test were to be carried out. The protocol was submitted for ethics approval by the Mount Kenya University Ethics and Scientific Board²². After ethics approval was done, a research license was sought and granted by the National Council for Science, Technology, and Innovations (NACOSTI)²³. Further entry to collect data in the selected Counties was provided by the County Health Departments of Health.

A 2-day training was organized for the enumerators at the M-Gas offices in Nairobi County on 28th and 29th October 2021. The training was facilitated by Lilian Kagume from ClimateCare (Project representative and CAD manager) in collaboration with Dr.

²² [Directorate of Research and Innovation \(mku.ac.ke\)](https://mku.ac.ke)

²³ <https://www.nacosti.go.ke>

J. Mwitari (an experienced public health specialist). The objectives of the training were seven-fold which were to enable the CHVs to:

1. Understand the Circle Gas Smart Meter LPG project
2. Understand the purpose of the baseline surveys and the Kitchen Performance tests
3. Be able to identify a potential household for baseline surveys and the Kitchen Performance tests
4. Be able to undertake baseline surveys and the Kitchen Performance tests
5. Understand and explain the role of an enumerator in survey data collection
6. Explain and demonstrate the qualities of a good enumerator
7. Understand the data collection instruments, and use of the Mobenzi data collection tool.

During the training, the enumerators were taken through the following areas:

Introduction to Circle Gas LPG Program

In this session, the participants were taken through the background, technology, implementation plan and benefits of the program activity. The M-Gas team demonstrated to the participants how the technology works, and its benefits compared to the traditional baseline stove.



Figure 8: An M-Gas representative taking the CHVs through a demo of how the smart meter technology works during day 1 training at M-Gas offices in Mukuru Kwa Njenga, Nairobi.

Source: Enumerators' training 2021

Principles of Effective Data Collection

This was an interactive session facilitated by Dr. James Mwitari, the participants were taken through the role of the enumerator in data collection, good enumerator skills and principles of effective data collection techniques.



Figure 9: Dr. Mwitari discussing with the participants the principles of good data collection.

Source: Enumerators' training 2021

Purpose of the surveys

Ms. Lilian Kagume from ClimateCare facilitated the session where the participants were taken through purpose and procedures of conducting the baseline surveys and Kitchen Performance Tests (KPTs).

Introduction to Mobenzi Data Collection Tool

The Mobenzi²⁴ data collection app was used for collecting data. The app provides a modern solution for collecting real time data and digitizing field work activities. During the training, the participants were taken through the Introduction, installation, use, benefits, and challenges of using the Mobenzi application. The app was installed on the enumerators smartphones and a practical session done on how to use the app during data collection. The management of the app during data collection was coordinated by an experienced specialist to ensure effective and quality data collection and control.

²⁴ [Mobenzi - Elegant technology enabling impact](#)



Figure 10: Participants practicing how to use the Mobenzi app during the training session.

Source: Enumerators' training 2021

Role plays

A practical session in form of role plays was held to allow the participants to reviewing the baseline survey and KPT procedures, practice using the survey questionnaire and the Mobenzi app.



Figure 11: Participants taking part in the role play exercise during the training.

Source: Enumerators' training 2021

b) Representativeness: The baseline survey requires in person interviews with a robust sample of end users without project technologies that are representative of end users targeted in the project activity.

The conducted in person interviews with a sample representative of the end-users targeted by the project activity. The baseline surveys were conducted within the applicable geographical areas covered under the project boundary. The surveys accounted for the different baseline scenarios that were identified by the project developer which included rural and urban populations.

The end-users surveyed were randomly selected and it was established that they were not using the project technology by the time of the survey.

The baseline survey was conducted in 14 counties. The selection of the 14 counties was informed by the 5-year distribution projections provided by the project developer. Since the project entails progressive installation of the units, the counties represent areas where the project intends to distribute units in the near future (5 years).

The sample size per county was then calculated as a proportion of the number of the projected units per county. There were a few counties which had only 1 selected sample and this was added to an adjacent county, or a county assumed to have similar baseline conditions. This is due to logistics challenge of travelling to the counties just to undertake 1 survey especially in the light of Covid 19 restrictions on movement.

A detailed procedure on the sample selection is provided in the Circle Gas Baseline and KPT Sampling spreadsheet provided as supporting evidence. A summary of the selected samples and the actual surveys conducted per County is provided below.

Table 13: Baseline survey sampling and surveys conducted per county

No	County	Projected installations	% Installations	Number of samples	Actual number sampled
1	NAIROBI	448,403	27.6%	36	36
2	KIAMBU	229,626	14.1%	18	18
3	NAKURU	200,727	12.4%	16	16
4	MOMBASA	112,889	6.9%	9	9
5	KAJIADO	110,451	6.8%	9	9
6	UASIN GISHU	106,595	6.6%	9	9
7	KISUMU	77,372	4.8%	6	6
8	TRANS NZOIA	59,915	3.7%	5	5
9	MIGORI	54,054	3.3%	4	4
10	NAROK	53,338	3.3%	4	4
11	HOMA BAY	46,584	2.9%	4	4
12	SIAYA	43,136	2.7%	3	3
13	LAIKIPIA	42,104	2.6%	3	3
14	BUSIA	39,606	2.4%	3	3
	Total	1,624,799		130	130

Source: Circle Gas Baseline and KPT Sampling spreadsheet tab Baseline survey sampling

In person interviews were held with the end-users to collect the required baseline data. As demonstrated in table 13, the required sample size was achieved on the surveys conducted.

Sample Sizing: The baseline survey should be carried out for each baseline scenario using representative and random sampling, following these guidelines for minimum sample size:

Group size	Minimum sample size
<300	30 or population size, whichever is smaller
300 to 1000	10% of group size
> 1000	100

Since the group size for the project activity is more than 1000, the minimum sample shall be 100 households without using the project technology. These will be randomly selected from the areas targeted by the project activity.

d) Data collected: Information on the following was gathered during the surveys:

1. Users follow up	i. Address or location ii. Mobile telephone number and/or landline telephone number (when possible)
2. End user characteristics	iii. Number of people served by baseline technology. iv. Typical baseline technology usage patterns and tasks (commercial, institutional, domestic, etc.)
3. Baseline technology and fuels	v. Types of baseline technologies used and estimated frequency. vi. Types of fuels used and estimated quantities. vii. Seasonal variations in baseline technology and fuel use viii. Sources of fuels; purchased, hand-collected, or etc., and prices paid, or effort made, e.g. walking distances, persons collecting, opportunity cost

Baseline scenario survey findings

1. End-user characteristics

The baseline survey was conducted with 130 households. Out of these, 11.5% were men while 88.5% were women. The average household size was calculated at 5.18. The details of all households surveyed are recorded under the Baseline Survey data spreadsheet tab Basic information.

Table 14: Analysis of the end-users surveyed

Analysis	Outcome	Percentage
Total number of surveys	130	
Number of Males	15	11.5%
Number of females	115	88.5%
Average number of people per household	5.18	

Source: Baseline Survey Data Tab Basic information

2. Baseline technology and fuels

As summarised in table 9, 62 households (47.7%) were using firewood for cooking as the primary fuel, followed by kerosene and charcoal at 26.9% and 25.4% respectively. The main source of the baseline fuel was identified as purchasing from vendors (82.9%), while 10.8% of the households indicated the collect from forests/woodlands.

Table 15: Analysis of primary fuel sources

Source of primary fuel	outcome	%
Collection	14	10.8%
Purchasing	116	89.2%
		100.0%

Source: Baseline Survey Data Tab Baseline Fuel Consumption

The predominant secondary fuel was charcoal at 54.5% followed by kerosene at 27.3% as shown in table 10. The main source of the secondary fuels was also purchasing from vendors (90.9%), while 9.1% of the households collected from forests/woodlands.

Table 16: Analysis of secondary fuel sources

Source of Secondary fuel	Outcome	Percentage
Collection	1	9.1%
Purchasing	10	90.9%
		100.0%

Source: Baseline Survey Data Tab Baseline Fuel Consumption

During the survey, the average fuel consumption and expenditure incurred in getting the baseline fuels was also assessed. For the primary fuel, the average consumption was 6.08 kgs and the average expenditure was KES102.39 (10\$).

Table 17: Average primary fuel consumption and expenditure

Consumption/expenditure	Outcome
Average primary fuel consumption per week (kgs)	6.08
Average primary fuel expenditure per week (Kshs.)	102.39

Source: Baseline Survey Data Tab Baseline Fuel Consumption

The average secondary fuel consumption was 1.45 kgs and the average expenditure was KES58.64(5.8\$).

Table 18: Average secondary fuel consumption and expenditure

Consumption/expenditure		
Average secondary fuel consumption per week (kgs)		1.45
Average secondary fuel expenditure per week (Kshs.)		58.64

Source: Baseline Survey Data Tab Baseline Fuel Consumption

On seasonal variation of the baseline fuel consumption, 59.2% spent less during dry seasons while 70.0% spent more during the rainy season. During holidays and weekends 82.3% and 83.1% of the respondents reported using more fuels respectively.

Table 19: Assessment of baseline fuel consumption based on seasonal variation

Seasonal variation in fuel consumption	Outcome	Percentage
Dry season		
More	14	10.8%
Less	77	59.2%
No change	39	30.0%
Rainy season		
More	91	70.0%
Less	10	7.7%
No change	29	22.3%
Holidays		
More	107	82.3%
Less	2	1.5%
No change	21	16.2%
Weekends		
More	108	83.1%
Less	3	2.3%

No change	19	14.6%
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Source: Baseline Survey Data Tab Baseline Fuel Consumption

The analysis of the primary baseline stove technologies is presented in table 11. The frequency of use of the stoves was rated as daily by 98.5% of the respondents, while 0.8% reported using the stoves 5 days a week and, 0.8% twice in a week.

Table 20: Frequency of use of the primary baseline stoves

Frequency of use of primary baseline stove	Outcome	Percentage
Daily	128	98.5%
5 days a week	1	0.8%
Twice a week	1	0.8%

Source: Baseline Survey Data Tab Baseline Fuel Consumption

The analysis of the primary baseline stove technologies is presented in table 12. It was determined that 90.9% of the households use the secondary stoves daily while 9.1% use them twice a week.

Table 21: Frequency of use of the secondary baseline stoves

Frequency of use of secondary baseline stove	Outcome	Percentage
Daily	10	90.9%
Twice a week	1	9.1%
		100.0%

Source: Baseline Survey Data Tab Baseline Fuel Consumption

The level of satisfaction in using the baseline stoves was assessed among the households surveyed. The majority at 45.4% expressed dissatisfaction in using the baseline stoves while 39.2% were partly satisfied. Only 2.3% were more than satisfied with their baseline stoves.

Table 22: Assessment of the level of satisfaction in using the baseline stoves.

Level of satisfaction in using baseline stove	Outcome	Percentage
More than satisfied	3	2.3%
Satisfied	17	13.1%
partly satisfied	51	39.2%
Not satisfied	59	45.4%
		100.0%

Source: Baseline Survey Data Tab Baseline Fuel Consumption

3. Baseline Sustainable development assessment

The baseline sustainable development scenario was also assessed during the survey. In assessment of SDG 5, time spent in getting the baseline fuel was evaluated. Majority of the households (50%) reported spending less than 30 minutes in getting the fuel, while 25.4% reported spending between 30-60 minutes. Only 24.6% spent more than 1 hour in getting fuel

Table 23: Analysis of the time spent by households getting the baseline fuel

Time spent getting fuels	Outcome	percentage
Less than 30 minutes	65	50.0%
30 minutes-1 hour	33	25.4%
More than 1 hour	32	24.6%

Source: Baseline Survey Data Tab Baseline Fuel Consumption

The time spent on cooking using the baseline fuels was also assessed where majority (56.2%) reported a cooking time of between 1-3 hours. 25.4% and 11.5% spent 3-5 hours and less than 1 cooking respectively. Only 6.9% spent more than 5 hours.

Table 24: Analysis of the average time spent in cooking using the baseline fuels

Time spent cooking	Outcome	percentage
Less than 1 hour	15	11.5%
1-3 hours	73	56.2%
3-5 hours	33	25.4%
More than 5 hours	9	6.9%

Source: Baseline Survey Data Tab Baseline Fuel Consumption

In assessing SDG 3, respondents were asked to rate the current level of indoor smoke within their households. A majority (67.7%) expressed the smoke level was high while 26.9% rated it as moderate. Only 5.4% reported low smoke levels in their households.

Table 25: Assessment of baseline indoor smoke levels

Air quality parameters	Outcome	Percentage
Level of smoke		
High	88	67.7%
Moderate	35	26.9%
Low	7	5.4%

Source: Baseline Survey Data Tab Baseline SD Assessment

For incidence of indoor air quality related illnesses, itchy eyes/eye problems and coughing were the predominant illnesses reported by respondents at 54.6% and 32.3% respectively.

Table 26: Analysis of air quality related illnesses incidences

Incidence of air quality related illnesses	Yes	No
Coughing	42	32.3%
Itchy eyes/eye problems	71	54.6%
Respiratory illnesses	17	13.1%

Source: Baseline Survey Data Tab Baseline SD Assessment

e) Baseline non-renewable biomass (NRB) assessment

The baseline scenario identified by the project activity assumes use of non-renewable biomass (firewood and charcoal) as the main baseline fuel among the targeted population. The fractional non-renewability of biomass hence needs to be established

using the latest CDM tool, "Tool 30: Calculation of the fraction of non-renewable biomass, version 3.0".

The value of f_{NRB} of 0.9447 for Kenya has been calculated since the value provided in the CDM registry is expired. The value of f_{NRB} shall be fixed ex-ante during the crediting period.

Determination of the Share of Non-Renewable Biomass

According to methodological TOOL 30 vers.3, par. 8, the following equation shall be used to calculate f_{NRB} :

$$f_{NRB,y} = \frac{NRB}{NRB+RB} \dots\dots\dots \text{(Equation 1)}$$

where:

$f_{NRB,y}$	Fraction of non-renewable biomass in the country (fraction or %)
NRB	Quantity of non-renewable biomass (t/yr)
RB	Quantity of renewable biomass (t/yr)

$$NRB = H - RB \dots\dots\dots \text{(Equation 2)}$$

Where:

H	=	Total consumption of woody biomass in the applicable area in the relevant period (tonnes), determined either through: (a) Official statistics or reports or peer-reviewed literature;
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$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other})) \dots \text{(Equation 3)}$$

Where	
$MAI_{forest,i}$	Mean Annual Increment of woody biomass growth per hectare in sub-category of forest areas (t/ha/yr)
$MAI_{other,i}$	Mean Annual Increment of woody biomass growth per hectare in sub-category of other wooded land areas (t/ha/yr)
$F_{forest,i}$	Extent of forest in sub-category i (ha)

$F_{\text{other},i}$	Extent of other wooded land in sub-category i(ha)
P_{forest}	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (ha)
P_{other}	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other wooded land areas (ha)
i	Sub-category i of forest areas and other wooded land areas

RB	t/yr	2,305,419
H	t/yr	41,700,666
NRB	t/yr	39,395,247
f_{NRB}		0.9447

Refer to the determination of NRB spreadsheet for detailed calculation.

f) Baseline firewood consumption and target population characteristics

The quantity of fuel consumed in the baseline scenario was determined based on the estimated average annual consumption of firewood, charcoal, and kerosene per household (tonnes/year), which was derived, in line with the section 4 of the applied methodology.

The Kitchen Performance Tests were carried to determine the baseline fuel consumption and saving values and the resultant result has been applied in the emission reduction calculations.

The KPT protocol

The Kitchen Performance Test Procedure stipulated under Annex 2 of the applicable methodology was applied. The primary objectives of the KPT were:

- a) To measure the quantity of the baseline fuel used.
- b) To demonstrate the fuel-saving quality of a Circle Gas Smart meter LPG unit in the household.

Kitchen Tests were carried out during the period between 02/11/2021 to 06/11/2021 among 81 randomly selected households.

The KT fuel measurement involved an independent test using the baseline stove over a period of 3 days. The fuel measurement for wood and charcoal was done using calibrated digital scale while kerosene was done using calibrated 1 Litre clear plastic measuring jugs. The calibration of the scales was done by an approved body and the calibration certificates have been submitted as supporting evidence (see *M-Gas calibration certificates*). The calibrated 1 Litre clear plastic measuring jugs were purchased in their calibrated form and there was no need for further calibration.



Figure 12: Kerosene measurement using a calibrated cup at Amina Mohammed's household in Mathare, Nairobi County.

Source: KPT Survey 2021



Figure 13: Fuel measurement using a digital scale at Kwa Kahindi-Fedson Kazungu household in Mombasa County.
Source: KPT Survey 2021

Procedure

1. Households were randomly selected to participate in the test. The households were derived from areas where the Circle Gas Smart meter LPG units have not been distributed.
2. An appropriate 3-day test period was chosen. The selection ensured that times where abnormal cooking activities are likely to occur such as festivals, holidays and weekends were avoided.
3. The enumerators explained to the selected households the purpose of the test and arranged to measure their fuel usage each day for a 3-day period. The households were encouraged not to change their cooking behaviour/patterns throughout the test period and not to give away any fuel to relatives/friends as this would lead to disqualification from the survey. The aim was to capture their usual behaviour in the kitchen, as if no tests were happening, to feed the usual variation of people with the usual variation of food types.
4. During day 1, the enumerators captured the household information and any other information which was deemed useful to the KT exercise. The enumerators then took a picture of the household's kitchen capturing the baseline stoves. This data was collected using the Mobenzi application.
5. While conducting the tests, the households were asked to use fuel only from the designated stock which had been pre-weighed. The enumerators visited the

households for 3 consecutive days to check that they were using only fuel from the weighed stock and are not adding un-weighed fuel to the stock. where more fuel was needed, this was weighed before adding it to the stock and the mass was added in the fuel data sheet.

6. The enumerators also inquired about the number of people being served each day, Number of meals they took, number of times the stoves were used and recorded if the household used any additional stoves.

7. At the end of the test, the final wood remaining was weighed and recorded. The fuel used for the test was provided by the project implementor to help control the weighing and use of fuel, rather than have the households use fuel they had bought themselves.

8. The project activity makes use of the smart meter technology which measures real time fuel consumption by all the end-users on a daily basis. Kitchen performance tests for the project technology were therefore not needed as this data is readily available.

KPT Sampling

The same approach used in determining the sample size for baseline surveys was applied in determining the KT sample size. The samples for the KT were derived from 14 counties where the project intends to progressively distribute the project technology in the near future. The number of samples per county was proportioned based on the projected installations of units in the counties as presented in table 27. A sample size of 81 households were randomly selected for the KT survey. These were randomly selected from the 130-sample selected for the baseline surveys.

Table 27: KT sample sizing

KPTs Sample sizing					
No	County	Projected installations	% Installations	Sampling	Actual KTs done
1	NAIROBI	448,403	27.6%	22	22
2	KIAMBU	229,626	14.1%	11	11
3	NAKURU	200,727	12.4%	10	10
4	MOMBASA	112,889	6.9%	6	6
5	KAJIADO	110,451	6.8%	5	5
6	UASIN GISHU	106,595	6.6%	5	5
7	KISUMU	77,372	4.8%	4	4

8	TRANS NZOIA	59,915	3.7%	3	3
9	MIGORI	54,054	3.3%	3	3
10	NAROK	53,338	3.3%	3	3
11	HOMA BAY	46,584	2.9%	2	2
12	SIAYA	43,136	2.7%	2	2
13	LAIKIPIA	42,104	2.6%	3	3
14	BUSIA	39,606	2.4%	2	2
	Total	1,624,799		81	81

Source: Circle Gas baseline surveys and KPT spreadsheet tab KPT sampling

Annex 2 of the applied methodology provides complementary guidelines for Kitchen Performance Testing. Since the KPTs were conducted for the baseline scenario, the sampling requirements provided in table 5 of the annex were followed. A co-efficient of 0.5 was identified and the sample required was 70. The extra samples were included to cater for any sample size attrition.

Table 5: Sample sizes in cases of SINGLE samples (where the tests are conducted for either baseline or project scenario but not both).

COV	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
90/10 precision	12	26	45	70	101	137	179	226	279

The sample size selected has been found to fulfil the required 90/10 precision as calculated using the Raosoft sample size calculator²⁵ as shown in figure below.

²⁵ [Sample Size Calculator by Raosoft, Inc.](#)



Sample size calculator

What margin of error can you accept? 5% is a common choice	10 %	The margin of error is the amount of error that you can tolerate. If 90% of respondents answer <i>yes</i> , while 10% answer <i>no</i> , you may be able to tolerate a larger amount of error than if the respondents are split 50-50 or 45-55. Lower margin of error requires a larger sample size.
What confidence level do you need? Typical choices are 90%, 95%, or 99%	90 %	The confidence level is the amount of uncertainty you can tolerate. Suppose that you have 20 yes-no questions in your survey. With a confidence level of 95%, you would expect that for one of the questions (1 in 20), the percentage of people who answer <i>yes</i> would be more than the margin of error away from the true answer. The true answer is the percentage you would get if you exhaustively interviewed everyone. Higher confidence level requires a larger sample size.
What is the population size? If you don't know, use 20000	1624799	How many people are there to choose your random sample from? The sample size doesn't change much for populations larger than 20,000.
What is the response distribution? Leave this as 50%	50 %	For each question, what do you expect the results will be? If the sample is skewed highly one way or the other, the population probably is, too. If you don't know, use 50%, which gives the largest sample size. See below under More information if this is confusing.
Your recommended sample size is	68	This is the minimum recommended size of your survey. If you create a sample of this many people and get responses from everyone, you're more likely to get a correct answer than you would from a large sample where only a small percentage of the sample responds to your survey.

Online surveys with Vovici have completion rates of 66%!

Alternate scenarios

With a sample size of	100	200	300	With a confidence level of	90	95	99
Your margin of error would be	8.22%	5.82%	4.75%	Your sample size would need to be	68	97	166

Save effort, save time. Conduct your survey online with Vovici.

More information

If 50% of all the people in a population of 20000 people drink coffee in the morning, and if you were repeat the survey of 377 people ("Did you drink coffee this morning?") many times, then 95% of the time, your survey would

KPT results

1. Baseline fuels

From the KT survey conducted 42/81 (51.9%) household were using firewood, followed by kerosene and charcoal at 24.7% and 23.5% respectively as the primary baseline fuels.

Table 28: Analysis of the primary baseline fuels used by households selected for KT survey

Primary Baseline fuels	outcome	%
Firewood	42	51.9%
Charcoal	19	23.5%
Kerosene	20	24.7%
		100.0%

Source: KPT data spreadsheet tab Combined KT data

Out of the 81 households, only 2 (2.5%) were using secondary fuels, both of which were using kerosene.

Table 29: Analysis of the secondary baseline fuels used by households selected for KT survey

Secondary Baseline fuels	outcome	%
Firewood	0	0.0%

Charcoal	0	0.0%
Kerosene	2	2.5%
		2.5%

Source: KPT data spreadsheet tab Combined KT data

2. Baseline stove technologies

The baseline technologies used for conducting the KTs are summarised in table 30. The majority (38.3%) were found to use traditional 3 stone fuelwood stove, followed by kerosene and non-improved charcoal stove at 24.7% and 23.5% respectively.

Table 30: Analysis of the primary baseline stoves technologies used by households

Primary Baseline stoves	Outcome	Percentage
Improved 3 stone fuelwood stove	11	13.6%
Traditional 3 stone fuelwood stove	31	38.3%
Kerosene stove	20	24.7%
Non-improved charcoal stove	19	23.5%
Total	81	100.0%

Source: KPT data spreadsheet tab Combined KT data

For the secondary stove technologies, the 2 households were using kerosene stoves for cooking.

Table 31: Analysis of the secondary baseline stoves technologies used by households

Secondary Baseline stoves	Outcome	Percentage
Improved 3 stone fuelwood stove	0	0.0%
Traditional 3 stone fuelwood stove	0	0.0%
Kerosene stove	2	100.0%
Non-improved charcoal stove	0	0.0%
Total	2	100.0%

Source: KPT data spreadsheet tab Combined KT data

3. Average household size

The number of people living in the household was assessed every day during the 3-day KPT survey. The average number of people per day was then determined, and the 3-day average calculated to be 5.45 as shown in table 32.

Table 32: Summary of average family size

Household size	outcome
Average family size Day 1	5.44
Average family size Day 2	5.42
Average family size Day 3	5.48
Average for 3 days	5.45

Source: KPT data spreadsheet tab Combined KT data

4. Meals served and cooked

The number of meals cooked and served by each household was assessed during each day of the KT. An average for each day was then determined, and the overall average calculated. The average number of meals served was found to be 2.74, while the average number of meals cooked was 2.64 as shown in tables 33 and 34.

Table 33: average meals served

Meals served	outcome
Average meals served Day 1	2.72
Average meals served Day 2	2.78
Average meals served Day 3	2.74
Average for 3 days	2.74

Source: KPT data spreadsheet tab Combined KT data

Table 34: average meals cooked

Number of Meals cooked	outcome
Average meals cooked Day 1	2.69
Average meals cooked Day 2	2.62
Average meals cooked Day 3	2.60
Average for 3 days	2.64

Source: KPT data spreadsheet tab Combined KT data

5. Fuel consumption

As indicated in table 28, KPTs were conducted to determine the amount of the fuels identified consumed by households. The fuel was weighed and recorded during the 3-day test period and the findings are as follows.

a. Firewood

42 households were using firewood. The average firewood consumption per day was determined, and the overall average for the 3-day calculated as presented in table 35. The average firewood consumption is 6.92 kgs per household per day.

Table 35: Average firewood consumption

Households using firewood	42
Average firewood consumption (kgs)	outcome
Average firewood consumption Day 1	7.94
Average firewood consumption Day 2	6.91
Average firewood consumption Day 3	5.91
Average for 3 days	6.92

Source: KPT data spreadsheet tab Firewood KT

b. Charcoal

19 households were using charcoal. The average charcoal consumption per day was determined, and the overall average for the 3-day calculated as presented in table 36. The average charcoal consumption is 1.72kgs per household per day.

Table 36: Average charcoal consumption

Households using charcoal	19
Average charcoal consumption (kgs)	outcome
Average charcoal consumption Day 1	2.42
Average charcoal consumption Day 2	1.42
Average charcoal consumption Day 3	1.31
Average for 3 days	1.72

Source: KPT data spreadsheet tab Charcoal KT

c. Kerosene

22 households were using kerosene for cooking in the baseline. The average kerosene consumption per day was determined, and the overall average for the 3-day calculated as presented in table 37. The average kerosene consumption is 0.76 litres per household per day.

Table 37: Average kerosene consumption

Households using kerosene	22
Average kerosene consumption (liters)	outcome
Average kerosene consumption Day 1	0.78
Average kerosene consumption Day 2	0.76
Average kerosene consumption Day 3	0.73
Average for 3 days	0.76

Source: KPT data spreadsheet tab Kerosene KT

g) Relevant applicable legislation and how effectively these are enforced (GS4GG Principle 1).

In the host county (Kenya), there is no requirement that households must use LPG, smart meter technology or LPG cookstoves for cooking or participate in any project which targets to implement an LPG cooking project. This is therefore a voluntary action by the project developer, and it is not being implemented as result of existing regulation requirement. The end-users will be provided with a complete LPG unit

comprising of a 13kg cylinder fitted with a smart meter, smart card, LPG cookstove and hose at no cost, and they will only be required to pay for the gas.

Kenya has a large population and fast-growing economy. However, the Kenya's energy sector remains underdeveloped, with the largest proportion of households relying on non-renewable biomass sources for cooking. As of 2017 (the most recent year for which comprehensive data are available), 93% of all residential and commercial energy consumption in Kenya was fueled by biomass sources, mainly charcoal and firewood.

Lack of access to modern energy for cooking is responsible for a high mortality and impaired quality of life for millions of people in low and middle-income countries like Kenya, who are exposed to hazardous smoke from cooking with biomass and kerosene.

In the project scenario, the project activity seeks to promote uptake of the metered LPG technology which will enable the end-users using the non-renewable biomass and kerosene fuels using inefficient cookstoves in the baseline to switch to LPG fuel and more efficient cookstoves for cooking.

B.5 Demonstration of additionality

Section 3.3.1 of the applied methodology, TPDDTEC Version 4.0 provides that "The project developer must show that the project could not or would not take place without the presence of carbon finance".

Further, the project developer shall demonstrate additionality prior to project registration by conforming to additionality requirements of one of the options below,

- a. Applicable GS4GG Activity Requirements.
- b. CDM Tool 01 - Tool for the Demonstration and Assessment of Additionality.
- c. CDM Tool 19- Demonstration of additionality of microscale project activities; (not applicable to Gold Standard microscale projects)
- d. CDM Tool 21 – Demonstration of additionality of small-scale project activities; (applicable to small-scale projects only)
- e. An approved Gold Standard VER additionality tool

Pursuant to section 4.1.9 of Community activity service requirements, "Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification".

- (a) Positive list
- (b) Projects located in LDC, SIDS, LLDC4
- (c) Microscale projects

Under Annex B on positive list, section 1.1.5 provides that "Project activities that involve technologies and/or practices providing thermal energy to the user that have less than 20% adoption rate among the target users. The most recent available data on adoption rates shall be used to demonstrate compliance with the 20 percent threshold in the target geographical area. In no case shall data be used if older than three years from the date of commencement of validation of the project activity".

The demonstration of additionality for the project activity has been done following this provision. The project activity entails distribution of LPG units to deliver thermal energy to households and can demonstrate that the project technology have less than 20% adoption rate in the target geographical area as discussed below.

The 2020 Kenya Continuous Household Survey Programme (KCHSP) 2020 Annual report conducted by the Kenya National Bureau of Statistics and published in 2021²⁶ report that LPG penetration in Kenya was at 13.9% in 2020. Previous findings from the the basic report for the Kenya Integrated Household Budget Survey²⁷ (2018), also conducted by the Kenya National Bureau of Statistics²⁸ had established that LPG access in the country in 2018 was at 13.4% as demonstrated in table 3.18 "Percentage Distribution of Households by Main Source of Cooking Fuel and Residence/ County" of the report. This includes 27.6% urban population and 2.5% rural population as summarized below.

²⁶ [Kenya - Kenya Continuous Household Survey Programme \(KCHSP\) - 2020 Annual \(knbs.or.ke\)](https://knbs.or.ke/kenya-kenya-continuous-household-survey-programme-kchsp-2020-annual)

²⁷ [Kenya Integrated Household Budget Survey - Kenya Integrated Household Survey - openAFRICA](#)

²⁸ [Home - Kenya National Bureau of Statistics, Nairobi, Kenya \(knbs.or.ke\)](https://knbs.or.ke/home)

3. HOUSING CONDITIONS AND AMENITIES

Table 3.18: Percentage Distribution of Households by Main Source of Cooking Fuel and Residence/ County

County/ Residence	Firewood	Electricity	Liquified Petroleum Gas (LPG)	Biogas	Kerosene	Charcoal	Straw Shrubs Grass	Animal Dung	Agricultural Crop Residue	Other	Number of Households ('000)
National.....	54.6	1.0	13.4	0.2	14.0	14.6	0.0	0.1	0.2	1.6	11,415
Rural.....	84.3	0.3	2.5	0.2	2.3	8.9	0.0	0.0	0.3	0.9	6,442
Urban.....	16.1	2.0	27.6	0.2	29.0	21.9	0.0	0.1	0.0	2.4	4,972

Figure 14: Percentage distribution of cooking fuels among households in Kenya

Source: Table 3.18 Basic Report: The Kenya Integrated Household Budget Survey (KIHBS,2018)

The Kenyan government has sought to encourage displacement of use of biomass and kerosene for cooking in households with LPG. Key among these is a policy goal of 35% of the population using LPG for cooking by 2030 to reduce reliance on charcoal and kerosene. Working through state-owned National Oil Corporation of Kenya (NOCK), the government began offering LPG cooking equipment (6 kg cylinder with ringtop burners) at a discount to poorer rural households in 2018 through a program called “Mwananchi Gas program”. However, the program was put on hold shortly before the formal launch (after a pilot in two counties) due to several operational and legal issues²⁹. In addition, retail prices of LPG are not regulated, so despite favorable policy environment and equipment subsidies, the cost of LPG reportedly remains out of reach for most households in Kenya.

In contrast to the policy goal, in 2020, a new Finance Bill was passed by Government, removing the VAT exemption for LPG. As of 1 July 2021, LPG was subjected to the standard 16% tax rate³⁰. This has pushed the price of LPG higher further limiting its affordability and adoption by households.

Another key access barrier to LPG in the host country is affordability. While LPG costs are frequently lower relative to purchased biomass fuels on a per kilogram basis, an unreliable supply of LPG and the financial outlay required to pay upfront for large units of gas (e.g., 6kg-15kg cylinders) under the standard Branded Cylinder Recirculation Model (BCRM) is prohibitive for many.

Further, among those that can afford, the high initial cost of purchasing the complete LPG unit, insecure incomes and precarious financial circumstances prevent some households from seeking full cylinder LPG refills, leading to its unsustainable usage as a primary cooking fuel. This consequently results to resumption in the use of the baseline fuels and inefficient baseline stove technologies.

²⁹ Sh3bn cooking gas for the poor hit by fraud - Business Daily (businessdailyafrica.com)

³⁰ <https://cleancooking.org/wp-content/uploads/2021/07/631-1-1.pdf>

It is critical to achieve year-round affordability of LPG to facilitate universal access to affordable, reliable, sustainable, and modern household energy. This can be attained using the innovative approach adopted by the project activity which seeks to provide a complete LPG unit to end-users at no cost and they only purchase the fuel, using the Pay-as-you-go (PAYG) model.

Pay-as-you-go (PAYG) is a consumer finance mechanism that potentially relieves the financial barrier to sustained clean energy access by allowing consumers to purchase LPG in small increments (via mobile banking). Using the smart meters enables consumers to track LPG consumption in real-time to ensure timely home deliveries of refills, which eliminates the need to travel to acquire fuel and minimizes safety risks from illegal cylinder refills associated with conventional LPG systems.

The Circle Gas Smart Meter LPG Program in Kenya is the first of its kind PAYGO enabled metered LPG project. The project will provide the complete units (LPG cylinder, cooker, and smart meter) to end users at zero cost or at a subsidized price for the stove, and the household will be required to pay for fuel. This will remove the financial barrier associated with upfront costs of acquiring an LPG unit. By enabling households to pay in small increments, pay-as-you-go (PAYG) LPG will help promote clean cooking, and support continued LPG use for cooking among end-users even through periods of economic downturn. However, all this cannot be possible without support from carbon financing due to the costs associated with developing, implementing, and maintaining the program.

The project is therefore deemed to be automatically additional as demonstrated below following the Gold Standard Community Activity Service Requirements, Annex B - Positive List.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).

As per GS4GG Community services activity requirements, Version 1.2, Paragraph 4.1.9, Projects that meet any of the following criteria are deemed additional and therefore are not required to prove Financial Additionality at the time of design certification:

- (a) Positive list (Annex B of this document)
- (b) Projects located in LDC, SIDS, LLDC
- (c) Microscale projects

Describe how the proposed project meets the criteria for deemed additionality.	The project activity meets criterion 1. Positive list, Annex B of the Community Services Activity Requirements. Section 1.1.5 provides that "Project activities that involve technologies and/or practices providing thermal energy to the user that has less than 20% adoption rate among the target users. The most recent available data on adoption rates shall be used to demonstrate compliance with the 20 per cent threshold in the target geographical area. In no case shall data be used if older than three years from the date of commencement of validation of the project activity". The PP has demonstrated above that the current LPG adoption rate is 13.9% as discussed in the 2020 Kenya Continuous Household Survey Programme (KCHSP) Annual report conducted by the Kenya National Bureau of Statistics. It can therefore be demonstrated that the project technology has an adoption rate of less than 20% of end-users within the project boundary.
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B.5.1 Prior Consideration

According to the Gold Standard's Principles and Requirements (version 1.2), Retroactive projects shall submit the required documents for preliminary review (time of first submission) within one year of the project start date. Retroactive Project submitted at a date later than one year from the project start date will not be eligible for Gold Standard certification.

The project start date is 02/09/2020, and PP submitted the required documents for preliminary review on 01/09/2021 which is within one year of the project start date. Therefore, the project is eligible for Gold Standard certification.

The milestones made by the project activity are as follows:

Project milestones

Table 38: Key project milestones

Timeline	Milestone
02/09/2020	Start date of the project (first installation of the LPG unit to end-user).
05/08/2021	Stakeholder consultation physical meeting
01/09/2021	The first submission date to GS for preliminary review
24/11/2021- 24/01/2022	Stakeholder feedback round
06/01/2022	Project listing
13/01/2022	VVB contacting for validation

B.5.2 Ongoing Financial Need

As discussed in section B.5 above, the Circle Gas Smart Meter LPG Program in Kenya plans to provide the complete units (13kg LPG cylinder, LPG cookstove, and smart meter) to end users at zero cost, or subsidised cost for the cookstove and the end-user will only be required to pay for fuel. This will remove the financial barrier associated with upfront costs of acquiring a complete LPG unit. By enabling end-users to pay in small increments, pay-as-you-go (PAYG) LPG will help promote clean cooking, and support continued LPG use even through periods of economic downturn. However, all this cannot be possible without support from carbon financing due to the costs associated with developing, implementing, and maintaining the program.

The carbon financing is intended to provide finance to investment in the production of the smart meters and procurement of the LPG cookstoves. The funds will also support the distribution of the units to end-users and support the hiring and training of the field sales and services teams.

Without carbon revenue, the sustainability of the project cannot be achieved, and the rate of expansion will be slow due to the associated costs, and most people might not be reached to benefit from the project. Carbon revenues will contribute to the sustainability of the proposed VPA by making the units more accessible to potential beneficiaries in Kenya.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	13.2 Integrate climate change measures into national policies, strategies, and planning.	Amount of GHGs emissions avoided or sequestered
8 Decent work and economic growth	8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.	Total number of employees earning above local minimum wage.
7 Affordable and clean energy	7.1 By 2030, ensure universal access to affordable, reliable, and modern energy services.	Proportion of population with primary reliance on clean fuels and technology.
5 Gender equality	5.4 Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate.	Average time saving associated with cooking time and fuel collection
3 Good health and well being	3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.	Proportion of users that perceive a reduction in smoke levels.

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

Methodological choices/approaches related to SDG 13

In line with the applied methodology TPDDTEC, version 4.0, emission reductions can be estimated by using the following formulas provided under method 3 for projects making use of fossil fuel in the project scenario.

As per the GS requirement, projects making use of fossil fuels should assume baseline and project fuel is the same. Hence, the overall GHG reductions achieved by the project activity in year y are calculated following method 3 as detailed below.

Method 3. Baseline and project fuel(s) are different and emission reductions are from fuel switch and efficiency gains.

This method is applied when the baseline fuel and the project fuel or energy source are different, and the emission factors are different. Since the project activity is introducing a fossil fuel (LPG), only emission reductions from efficiency improvement shall be eligible, i.e. the baseline and project fuel emission factor shall be assumed to be the same.

The GHG reductions achieved by the project activity in year y shall be calculated as follows:

Baseline emissions

$$BE_y = \sum (b,p) N_{b,p,y} \times U_{p,y} \times (f_{NRB,b,y} \times SE_{b,y,CO2} + SE_{b,y,nonCO2}) \dots \text{(Equation 3)}$$

Where:

Parameter	Description	Value applied
BE_y	Baseline emissions for total project activity in year y (tCO2e/yr)	1.919
$\sum_{b,p}$	Sum over all relevant (baseline b/project p) pairs	1
$N_{b,p,y}$	Number of project technology-days included in the project database for baseline b/project p pair in year y (days)	365
$U_{p,y}$	Cumulative Usage rate for technologies in project scenario p in year y (fraction)	75%
$SE_{b,y,CO2}$	Specific CO2 emissions for a baseline b technology in year y (tCO2/technology*day)	0.00629

$SE_{b,y,non-CO2}$	Specific non-CO2 emissions for a baseline b technology in year y (tCO2e/technology*day)	0.00107
$f_{NRB,b,y}$	Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass (fraction). This term is included when the baseline practice was use of biomass, even if a fossil fuel baseline emission factor must be applied due to the project activity introducing fossil fuel.	0.9447

Project emissions

Project emissions are calculated as follows:

$$PE_y = \sum (b_{p,y} N_{b,p,y} \times U_{p,y} \times (f_{NRB,b,y} \times SE_{p,y,CO2} + SE_{p,y,nonCO2})) \dots \dots (Equation 4)$$

Where:

Parameter	Description	Value applied
PE_y	Project emissions for total project activity in year y (tCO2e/yr)	0.0333
$SE_{p,y,CO2}$	Specific CO2 emissions for a project p technology in year y (tCO2/technology*day)	0.000101
$SE_{p,y,non-CO2}$	Specific non-CO2 emissions for a project p technology in year y (tCO2e/technology*day)	0.000026
$f_{NRB,b,y}$	Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass (fraction). This term is excluded from the equation when the project is using a fossil fuel.	0.9447

Assessment of Leakage emissions

Section 3.11 of the applied methodology outlines two options for assessment of leakage.

Option 1: Apply a default adjustment factor of 0.95 to the emission reductions to approximate leakage emissions. In this case, the term " $-\sum LE_{p,y}$ " in equations 1, 2 or 3 changes to " $* 95\%$ ".

Option 2: The project developer must evaluate the following potential sources of leakage and provide an evidence-based description and preliminary quantification of each potential source and its relevance for the project. This option has been chosen for the project activity as demonstrated below:

a. The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or with a higher intensity than would have occurred in the absence of the project.

The project boundary is defined as the location of the installed metered LPG cooking technology and the environs within the target areas where the project technologies are utilized. The target area for this project is defined by the counties in which the project activity is to be implemented within the Republic of Kenya. As demonstrated from the findings of the baseline surveys conducted, within the project boundary, the predominant baseline technologies are inefficient stoves (firewood and charcoal) which are mainly restricted to domestic setting. The purpose of the project activity is to accelerate uptake of the metered LPG cooking technology among end-users using the inefficient baseline stoves. It is highly unlikely that these technologies will be reused outside the target area as preference of cooking technologies vary across households.

b. Members of the population who do not participate in the project, and previously used lower emitting energy sources, instead use the non-renewable biomass or fossil fuels saved under the project activity.

The baseline fuels used by the end-users were found to be sourced by purchasing from vendors or obtained individually from forests/woodlands. Domestic fuel consumption is a function of market forces of demand and supply. Participation into the project activity by the end-users is voluntary and there is no indication that the fuel consumption would increase amongst non-project users who previously relied on lower-emitting sources.

c. The project significantly reduces the NRB fraction within an area where other GHG mitigation project activities account for NRB fraction in their baseline scenario.

The project is anticipated to have minimal aggregate effect on the total NRB fraction in the country in comparison to other CDM or VER project activities in the baseline scenario.

d. The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of space heating or by retaining some use of inefficient technology.

In the targeted project area, the average evening temperature is about 22°C and people do not use any space heating technology. Therefore, there is no anticipated loss of space heating because of using the project technology for cooking. The project only displaces the practice of using non-renewable biomass and kerosene and use of inefficient baseline stoves for cooking. Hence, there is no anticipated leakage as end-users in the target area do not light fires for space heating.

e. By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution with this technology by households who commonly used a technology with relatively lower emissions.

The cooking technology being promoted by the project activity seeks to accelerate uptake of metered LPG cooking technology among end-users. Majority of the households in the targeted project areas use highly emitting fuels such as firewood, charcoal, and kerosene for cooking, using inefficient baseline stove technologies. LPG is a clean fuel with very minimal emissions. There is no envisioned scenario where these end-users will substitute technologies with relatively lower emissions technology with lower emissions because of promotion of the project technology.

From this assessment, leakage risks of the project activity are deemed very low and can be ignored. Leakage emissions (LE_y)=0 tCO₂e.

Leakage investigation shall be conducted every two years. The leakage emissions parameter has been included in the monitoring plan.

Emission reductions

$$ER_y = BE_y - PE_y - \sum LE_{p,y} \dots \dots \dots \text{(Equation 5)}$$

Where:

Parameter	Description	Value applied
ER_y	Emission reduction for total project activity in year y (tCO ₂ e/yr)	1.886
BE_y	Baseline emissions for total project activity in year y (tCO ₂ e/yr)	1.919
PE_y	Project emissions for total project activity in year y (tCO ₂ e/yr)	0.0333
$LE_{p,y}$	Leakage for project scenario p in year y (tCO ₂ e/yr)	0

Methodological choices/approaches related to SDG 3

Parameter: Proportion of users that perceive a reduction in smoke levels.

The VPA aims to install metered LPG cooking technology which will replace the use of non-renewable biomass baseline fuels such as firewood and charcoal and kerosene with LPG for cooking. Use of these baseline fuels emits a lot of smoke and other air pollutants, which will be reduced by using the project technology.

The proportion of users that perceive a reduction in smoke levels will be monitored through a qualitative question in the monitoring survey form. The parameter will then be analyzed and expressed as a percentage based on responses given.

The project activity seeks to achieve 100% improvement in indoor air quality through reduced smoke levels among end-users using the project technology.

Methodological choices/approaches related to SDG 5

Parameter: Average time saving associated with cooking time and fuel collection

The project activity targets to provide the complete LPG unit to end-users, who in return will be required to pay for the fuel they use. The LPG cylinder refill and maintenance of the project technology will be done by the VPA implementor at no extra cost. This will enable more end-users and particularly women to benefit from the project technology to save on time spent in searching for and collecting fuel, and in cooking. The time saved can be spent on other income generating activities. Using the project technology is expected to lead to significant time saving associated with cooking time and fuel collection compared to use of the baseline fuels.

The parameter will be monitored through a qualitative question in the monitoring survey form. This will then be compared with the average time associated with cooking time and fuel collection in the baseline, which will be achieved by conducting the baseline surveys.

Average time saving associated with cooking time and fuel collection

= average time spent in cooking time and fuel collection in the baseline scenario-
Average time saving associated with cooking time and fuel collection in the project scenario.

Using the project technology is expected to lead to time savings of about 3.2 hours per day.

Methodological choices/approaches related to SDG 7

Parameter: Proportion of population with primary reliance on clean fuels and technology

This parameter will be monitored based on the actual number of operational units installed as indicated in the distribution database. The project developer shall maintain an updated distribution database for all the units installed by the project.

The project is expected to distribute about 1,928,113 complete LPG units among end-users within the project boundary.

Methodological choices/approaches related to SDG 8

Parameter: Total number of employees earning above local minimum wage.

The project activity will lead to job creation, and this parameter will be monitored to determine the number of people employed by the project and paid within the national minimum wage for the host country. This parameter will be monitored directly by reviewing the employment data which will be kept by the project implementer. The data shall include but not limited the number of people employed and their salaries/wages, and their gender and age.

In the baseline there were no people employed by the project. In the project scenario, the project targets to directly employ 1,000 people each year in the distribution, installation, sales, and maintenance functions of the project.

B.6.2. Data and parameters fixed ex ante

SDG13

Data/parameter	SDG 13 ICS 1: Baseline scenario survey results
Unit	N/A
Description	Report of the results of the baseline scenario survey
Source of data	The report presents the results of the Baseline Scenario Survey
Value(s) applied	The baseline scenario findings are presented in section B.4 of the VPA-DD and the values are presented on tables 9 to 37.
Choice of data or Measurement methods and procedures	The applied data have been informed by the findings of the baseline survey which was conducted with 130 households
Purpose of data	Calculation of baseline scenario
Additional comment	A baseline survey was undertaken as detailed in section B.4. The surveys were done on 130 randomly selected who were not using the project technology. The survey was undertaken between 01/11/2021 to 03/11/2021, by a group of 27 trained enumerators derived from Community Health Volunteers (CHVs) from the respective county health departments where baseline surveys were done. The finding obtained have been reviewed and has been found to be representative of the baseline scenario. These will be applied during the crediting period.

Data/parameter	SDG 13 ICS 2: Project technology description
Unit	N/A
Description	The detailed description of the project technology shall include as a minimum: - Manufacturer name, - product name (if applicable), - technology type, - capacity characteristics, - continuous useful energy output demonstration, - rated thermal efficiency, for the following cases: - improved biomass cookstoves, fuel-based ovens, water heaters, improved fossil fuel cookstoves (solid fuel or existing liquid & gaseous fuel projects), and renewable fuel fired stoves.³¹ - Any performance certifications from National Standards body or certification body recognized by national standards body also shall be provided.
Source of data	The methodology requires that any of the following sources shall be used: - Manufacturer specifications - Certifications by national standards body or an appropriate certification party recognized by national standards body - Commercial guarantee - Technical reports from the installer - For stoves built on-site at the end user location, reports of Standard WBT by stove manufacturer or installer Professional opinion or expert opinion is not accepted as a source for this parameter. The Project technology description has been made based on manufacturer specifications.
Value(s) applied	The PP has sourced the data for the parameter from manufacturer specifications.

³¹ Not applicable for the project since the project is not introducing improved cookstoves

	Smart meters - Manufacturer name: KopaTech - Product name: KopaMeter - Technology type: LPG Smart Meter - Capacity characteristics: KOPAMETER is a consumer grade Smart Meter for LPG (liquid petroleum gas) dispensed in small size canisters (up to 200 L capacity). - Continuous useful energy output demonstration: N/A - Rated thermal efficiency: N/A Cookstoves - Manufacturer name: Different based on brand - Product name: LPG cookstoves - Technology type: single or double burners -Gas consumption: Different based on brand but approximated at 0.5 kg/h (double), 0.27kg/h (single) -Dimension (H*W*D): Different based on brand but approximated 120*720*390 (double), 140*300*390 (single) -Weight: Different based on brand but approximated at 7kg (double burner), and 4kg (single burner) - Gas inlet: Inner diameter approximated at 9.5 mm - Continuous useful energy output: 5.65kW -Fuel type: LPG -Rated thermal efficiency: 50-60% The smart meter technology has received the Eurocert SA certification and has been certified to comply with essential health and safety requirements relating to the design and construction of products intended for use in potentially explosive atmospheres. The certificate has been provided as supporting evidence. The smart meter has also obtained a calibration certification from the Kenya Bureau of Standards (KEBS) which is the National standard body in the host country. The certificate confirms that the smart meter has achieved the required accuracy test (<i>See M-Gas KEBS certificate provided as supporting evidence</i>).
Choice of data or Measurement methods and procedures	Circle Gas Limited is the manufacturer of the project technology, and the manufacturers specifications provide the detailed description of the product technology.
Purpose of data	Description of the project technology
Additional comment	The smart meter is fixed on the LPG cylinder and only authorised representatives can remove it from the cylinder

to maximise on safety. The smart meter manual and stove manual detailing its manufacturer specifications has been provided as a support document. The project is not producing its own efficient LPG cookstove, the stoves used are the conventional brands in the market and will hence have different manufacturers.

Data/parameter	SDG 13 ICS 3: Expected technical life of project technology
Unit	Years
Description	The expected technical life of an individual project technology shall be defined in the VPA-DD.
Source of data	The methodology provides that any of the following sources shall be used: - Manufacturer specifications - Certification by national standards body or an appropriate certification party recognized by national standards body - Commercial guarantee or Guarantee from the installer - For stoves built on-site at the end user location, field reports, which comply with the general requirements for sampling, of average technical life of the same stove type operated under similar conditions (socioeconomic and cultural). Simulation modelling may be applied together with such field reports to estimate the average technical life. Professional opinion or expert opinion is not accepted as a source for this parameter.
Value(s) applied	7 years
Choice of data or Measurement methods and procedures	The PP has sourced the data for the parameter from manufacturer specifications.
Purpose of data	Determining the technical life of the project technology
Additional comment	The expected technical life of project technology is longer than the crediting period and therefore measures to ensure that end users are provided replacement technology of comparable or higher quality at the end of the technical life, by either replacing with comparable or better technology, or retrofitting essential parts with performance guarantee is not needed. Further, the project developer shall conduct continuous monitoring and maintenance of the units to ensure they are full operational during the crediting period. The smart meter manual detailing its manufacturer specifications has been provided as a support document.

Data/parameter	SDG 3 ICS 4: Indoor air pollution (IAP) levels of the project technology
Unit	NA
Description	For projects where cooking will move from outdoor to indoor or where the project technology reduces ventilation (for example, changing from a stove with chimney to improved stove with no chimney), demonstration that Indoor air pollution (IAP) levels are not worsened in the project scenario compared to the baseline, including PM 2.5 and carbon monoxide (CO) emissions.
Source of data	The methodology provides that for the description of IAP level of the project technology, any of the following sources shall be used: - Certification resulting from a manufacturer's test - Report of field testing of the technology - Report of lab testing of the technology - Results of modelling of the technology's operation under field conditions. - For stoves built on-site at the end user location, existing reports of lab or field testing of similar technology. Professional opinion or expert opinion is not accepted as a source for this parameter. For the IAP level of the baseline scenario, the following sources shall be used: - Certification resulting from of a manufacturer's test, - Report of field testing of the technology, - Report of lab testing of the technology, - Results of modelling of the technology's operation under field conditions, or - Expert opinion For both project and baseline technologies, if none of these are available, reference from published literature or report by independent agencies may be used as evidence, provided it is not more than 5 years old.
Value(s) applied	The findings of the referenced research done shows that the average PM2.5 exposures of LPG (project IAP) was measured to be 43 g/m3 compared to charcoal 115 g/m3

	and firewood 225 g/m ³ (baseline IAP) (see <i>table 3 on page 10 of the attached report</i> ”).
Choice of data or Measurement methods and procedures	The values applied have been informed by a published literature “Modelling Approaches and Performance for Estimating Personal Exposure to Household Air Pollution: A case Study in Kenya ³² ” conducted by Johnson et.al in 2020.
Purpose of data	Assessing the indoor air pollution (IAP) levels of the project technology
Additional comment	The technology being promoted by the project will promote a shift from outdoor to indoor cooking in some instances. Some households especially those using the traditional 3 stone firewood stoves were found to cook outdoors due to high level of smoke associated with the stove technology. The project activity is introducing use of LPG for cooking which a clean energy and less emitting compared to the identified baseline fuels. As evident from the study done the project technology is expected to significantly reduce IAP levels among the beneficiary end-users. The study report “Modelling Approaches and Performance for Estimating Personal Exposure to Household Air Pollution: A case Study in Kenya” done in Kenya by Johnson et.al and the Berkeley Air Monitoring Group in 2020 has been provided as a support document (see Johnson et al_Indoor Air report Mar21).

³² [Modeling approaches and performance for estimating personal exposure to household air pollution: A case study in Kenya - PubMed \(nih.gov\)](#)

Data/parameter	SDG 13 ICS 5: Avoidance of double counting or double claiming among project participants
Unit	NA
Description	Evidence of avoidance of double counting or double claiming with other parties directly involved with the project or programme.
Source of data	Written assertions with the project developer of the ownership rights and intention of selling the emission reductions resulting from the project activity directed at or signed with all the applicable parties of the following: - All other project participants - Project technology producers; and - Retailers of the project technology or the renewable fuel.
Value(s) applied	The PP has provided proof that the procedure for the carbon rights transfer with end-users of the project technology is in place.
Choice of data or Measurement methods and procedures	The mechanism for the carbon rights transfer has been put in place and the same shall be verified to ensure there is no double counting or claiming by the project participants.
Purpose of data	To demonstrate avoidance of double counting or double claiming among project participants
Additional comment	The project developer is the proprietor of the project technology and is in-charge of the distribution. No external third parties are involved in the distribution and hence the proof of carbon rights transfer has been demonstrated for the end-users of the project technology. Each end-user shall be required to consent/sign the end-user contract which stipulates the carbon rights transfer during installation of the units (<i>see Circle Gas_End user contract_confidential document provided as evidence</i>)

Data/parameter	SDG 13 ICS 6: Avoidance of double counting or double claiming with other mitigation actions
Unit	NA
Description	Review and analysis of mitigation actions in other voluntary market or UNFCCC/compliance mechanisms
Source of data	Using publicly available information from Gold Standard and other voluntary standards, at a minimum Verra and any recognized national or regional standards in the project location, and UNFCCC CDM project & PoA database: - Identify and list any mitigation actions of similar technology, i.e., that provide the same kind of output and use the same kind of equipment or conversion process, operating in overlapping spatial boundaries. If one or more are identified: - Describe the practices that will be implemented to ensure that the programme or project activity quantifies emission reductions only from technology it has implemented, - Describe the practices to avoid that the programme or project activity implementation displaces technology of other mitigation actions, and - Design a method to discount emission reductions in case the programme or project activity is found to displace or operate alongside another mitigation action.
Value(s) applied	None, no mitigation actions of similar technology, i.e., that provide the same kind of output and use the same kind of equipment or conversion process was found to operate in the project activities spatial boundaries.
Choice of data or Measurement methods and procedures	Review of publicly available information from Gold Standard and other voluntary standards
Purpose of data	To demonstrate avoidance of double counting or double claiming with other mitigation actions

Additional comment

The PP conducted a search on the CDM registry, Gold Standard and Verra registries via the following links:

[GSF Registry \(goldstandard.org\)](https://goldstandard.org/)

[CDM: Project Activities \(unfccc.int\)](https://unfccc.int/)

[CDM: Programme of Activities \(unfccc.int\)](https://unfccc.int/)

[Verra Search Page](#)

The PP did not identify any mitigation actions of similar technology, i.e. that provide the same kind of output and use the same kind of equipment within the project boundary. The project developer has also signed a commitment letter confirming the project is neither registered nor seeking registration with other carbon standards (*see Circle Gas registration with other programs letter provided*).

Data/parameter	SDG 13 ICS 7: Regulatory framework for provision of thermal energy services
Unit	NA
Description	Evidence that the project does not undermine or conflict with any national, sub-national or local regulations or guidance for thermal energy supply/devices or fuel supply or use
Source of data	List and provide a summary of any national, sub-national and local regulations or guidance for provision of thermal energy services/devices of the type of the project provides in the project boundary, including any tariff requirements. Describe how the project complies with the regulatory framework.
Value(s) applied	None
Choice of data or Measurement methods and procedures	The PP reviewed the regulatory framework for provision of thermal energy services within the host country (Kenya). From the assessment, it has been demonstrated that the project activity does not undermine nor conflict with any regulation or guidance relating to thermal energy supply/devices or fuel supply or use. A detailed analysis is provided in Table 2 and 3 in section A.1 of the VPA-DD.
Purpose of data	To demonstrate that the project does not undermine or conflict with any national, sub-national or local regulations or guidance for thermal energy supply/devices or fuel supply or use
Additional comment	The assessment has been undertaken at the start of the crediting period and shall be updated during the renewal of the crediting period.

Data/parameter	SDG 13 ICS 8: EF_{b,f,CO_2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of fuels in baseline scenario
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default, 112 tCO₂/TJ Charcoal: Methodology default, 112 tCO₂/TJ (combustion only) Methodology default, 165.22 tCO₂/TJ (includes charcoal production emissions) Methodology cap, 197.15 tCO₂/TJ (includes charcoal production emissions) Other fuels: IPCC defaults When emissions from fuel production, transport, and similar are included to determine a project-specific emission factor, then the following shall apply as well: - The project boundary must include these processes - Avoidance of double counting considerations must cover all steps in the project boundary. - The determination of the specific emissions from these sources is fully documented and evidenced in the PDD These provisions may be applied to include the actual GHG emissions happening upstream in charcoal production in the charcoal emission factor; however, emission factors higher than the methodology cap are not permitted.
Value(s) applied	$EF_{b,firewood,CO_2} = 112 \text{ tCO}_2/\text{TJ}$; $EF_{b,charcoal,CO_2} = 165.22 \text{ tCO}_2/\text{TJ}$ $EF_{b,kerosene,CO_2} = 71.9 \text{ tCO}_2/\text{TJ}$ $EF_{b,LPG,CO_2} = 63.1 \text{ tCO}_2/\text{TJ}$
Choice of data or Measurement methods and procedures	The most recent IPCC default values are applied for kerosene and LPG fuels as provided under IPCC 2006 (Vol 2, Chap 1, Table 1.4). The methodology default values of 165.22 tCO₂/TJ and 112 tCO₂/TJ are applied for charcoal and firewood respectively.
Purpose of data	Calculation of parameter SE_{b,y,CO_2} applied in baseline

	emissions estimation.
Additional comment	The charcoal emission factor includes the charcoal production emissions.

Data/parameter	SDG 13 ICS 9: EF_{b,f,nonCO2}
Unit	tCO ₂ /TJ
Description	Non-CO2 emission factor arising from use of fuels in baseline scenario
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default: - 9.46 tCO₂e/TJ (AR5 GWP) or 8.692 tCO₂e/TJ (AR4 GWP) Charcoal: Methodology defaults: - 5.865 tCO₂e/TJ (AR5 GWP) (combustion only) - 44.83 tCO₂e/TJ (AR5 GWP) (includes charcoal production emissions), - Methodology cap: 92.29 tCO₂e/TJ (AR5 GWP) (includes charcoal production emissions) – or - 5.298 tCO₂e/TJ (AR4 GWP) (combustion only) - 40.26 tCO₂e/TJ (AR4 GWP) (includes charcoal production emissions) - Methodology cap: 82.90 tCO₂e/TJ (AR4 GWP) (includes charcoal production emissions) Other fuels: Any of the following, in order of preference: - IPCC defaults - Project-specific field tests prior to first verification by a qualified entity that is certified or accredited by National Standards body - Project-relevant measurement reports by qualified entities - National defaults - Credible published literature for the project area If either project-specific or project-relevant results are used, these must be cross-checked with IPCC defaults and differences shall be justified using evidence. When emissions from fuel production, transport, and similar are included to determine a project-specific emission factor, then the following shall apply as well: - The project boundary must include these processes - Avoidance of double counting considerations must cover all steps in the project boundary - The determination of the specific emissions from these sources is fully documented and evidenced in the PDD

	These provisions may be applied to include the actual GHG emissions happening upstream in charcoal production in the charcoal emission factor; however, emission factors higher than the methodology cap are not permitted.
Value(s) applied	$EF_{\text{firewood,nonCO}_2} = 9.46 \text{ tCO}_2\text{e/TJ (AR5 GWP)}$; $EF_{\text{b,charcoal,nonCO}_2} = 44.83 \text{ tCO}_2\text{e/TJ (AR5 GWP)}$ $EF_{\text{b,kerosene,nonCO}_2} = 0.76 \text{ tCO}_2\text{/TJ}$ $EF_{\text{b,LPG,nonCO}_2} = 0.89 \text{ tCO}_2\text{/TJ}$
Choice of data or Measurement methods and procedures	For Firewood and Charcoal, the methodology AR5 GWP defaults are applied. For Kerosene and LPG, the IPCC 2006 (Vol 2, Chap 2, Table 2.9) data have been applied the values have been multiplied with the latest AR5 GWP data (GWPC _{H4} = 28 and GWP N ₂ O= 265)
Purpose of data	Calculation of parameter $SE_{b,y,\text{non-CO}_2}$ applied in calculation of baseline emissions.
Additional comment	The charcoal emission factor includes the charcoal production emissions.

Data/parameter	SDG 13 ICS 10: EF_{p,f,CO_2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of fuels in project scenario
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default, 112 tCO₂/TJ Charcoal: Methodology default, 112 tCO₂/TJ (combustion only) Methodology default, 165.22 tCO₂/TJ (includes charcoal production emissions) Methodology cap, 197.15 tCO₂/TJ (includes charcoal production emissions) Other fuels: - IPCC defaults
Value(s) applied	$EF_{p,firewood,CO_2} = 112 \text{ tCO}_2/\text{TJ}$ $EF_{p,LPG,CO_2} = 63.1 \text{ tCO}_2/\text{TJ}$; $EF_{p,charcoal,CO_2} = 165.22 \text{ tCO}_2/\text{TJ}$; $EF_{p,kerosene,CO_2} = 71.9 \text{ tCO}_2/\text{TJ}$
Choice of data or Measurement methods and procedures	For Firewood and Charcoal, the methodology defaults are applied. For Kerosene and LPG, the IPCC 2006 (Vol 2, Chap 1, Table 1.4) data have been applied.
Purpose of data	Calculation of parameter SE_{p,y,CO_2} applied in baseline emission calculations.
Additional comment	The charcoal emission factor includes the charcoal production emissions.

Data/parameter	SDG 13 ICS 11: $EF_{p,f, nonCO2}$
Unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor arising from use of fuels in project scenario
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default: - 9.46 tCO₂e/TJ (AR5 GWP) or - 8.692 tCO₂e/TJ (AR4 GWP) Charcoal: Methodology defaults: - 5.865 tCO₂e/TJ (AR5 GWP) (combustion only) - 44.83 tCO₂e/TJ (AR5 GWP) (includes charcoal production emissions), - Methodology cap: 92.29 tCO₂e/TJ (AR5 GWP) (includes charcoal production emissions) - or - 5.298 tCO₂e/TJ (AR4 GWP) (combustion only) - 40.26 tCO₂e/TJ (AR4 GWP) (includes charcoal production emissions) - Methodology cap: 82.90 tCO₂e/TJ (AR4 GWP) (includes charcoal production emissions) Other fuels: Any of the following, in order of preference: - IPCC defaults, - Project-specific field tests prior to first verification by a qualified entity that is certified or accredited by National Standard body - Project-relevant measurement reports by qualified entities, - National defaults

	- Credible published literature for the project area If either project-specific or project-relevant results are used, these must be cross-checked with IPCC defaults and differences shall be justified using evidence.
Value(s) applied	$EF_{\text{firewood,nonCO}_2} = 9.46 \text{ tCO}_2\text{e/TJ}$ (AR5 GWP) $EF_{\text{b,charcoal,nonCO}_2} = 44.83 \text{ tCO}_2\text{e/TJ}$ (AR5 GWP) ($EF_{\text{b,kerosene,nonCO}_2} = 0.76 \text{ tCO}_2\text{/TJ}$) $EF_{\text{LPG,nonCO}_2} = 16.10 \text{ tCO}_2\text{e/TJ}$
Choice of data or Measurement methods and procedures	For Firewood and Charcoal, the AR5 GWP methodology defaults are applied. For Kerosene and LPG, the IPCC 2006 Vol2 Chap 2 Table 2.9 and latest GWP of CH₄ and N₂O (WGI AR5 2013 Table 8A.1 p.8-88) values are applied. Because of the Gold Standard Foundation overall applicability criteria limiting the scope of fossil fuel switching projects to their energy efficiency component, the LPG Emission Factor value used here is a weighted average of the baseline fuels that are going to be displaced by the introduction of the LPG units. This will lead to a very conservative estimate of the actual emission reduction.
Purpose of data	To calculate parameter $SE_{p,y,\text{non-CO}_2}$ applied in calculation of baseline emission reductions.
Additional comment	The charcoal emission factor includes the charcoal production emissions.

Data/parameter	SDG 13 ICS 12: $NCV_{b,fuel}$
Unit	TJ/ton
Description	Net calorific value of the fuels used in the baseline
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default, 0.0156 TJ/ton Charcoal: Methodology default, 0.0295 TJ/ton Other fuels: - IPCC defaults - Project-specific testing by a qualified entity that is certified or accredited - Project-relevant measurement reports by a qualified entity If either project-specific or project-relevant results are used, these must be cross-checked with IPCC defaults and differences shall be justified using evidence. IPCC defaults, project-relevant measurement reports, or project-specific testing
Value(s) applied	$NCV_{b,firewood} = 0.0156$ TJ/ton $NCV_{b,charcoal} = 0.0295$ TJ/ton $NCV_{b,kerosene} = 0.0438$ TJ/ton $NCV_{b,LPG} = 0.0473$ TJ/ton
Choice of data or Measurement methods and procedures	For Firewood and Charcoal, the methodology defaults are applied. For Kerosene and LPG, the IPCC 2006 Vol2 Chap 1 Table 1.2 values are applied.
Purpose of data	To calculate parameter $SE_{b,y,CO2}$ and $SE_{b,y,non-CO2}$
Additional comment	The methodology default emission factor has been applied since methodology default NCV is applied.

Data/parameter	SDG 13 ICS 13: $NCV_{p,fuel}$
Unit	TJ/ton
Description	Net calorific value of the fuels used in the project
Source of data	The methodology provides that the following shall be applied: Wood: Methodology default, 0.0156 TJ/ton Charcoal: Methodology default, 0.0295 TJ/ton Other fuels: - IPCC defaults - Project-specific testing by a qualified entity that is certified or accredited - Project-relevant measurement reports by a qualified entity If either project-specific or project-relevant results are used, these must be cross-checked with IPCC defaults and differences shall be justified using evidence.
Value(s) applied	$NCV_{p,firewood} = 0.0156$ TJ/ton $NCV_{p,charcoal} = 0.0295$ TJ/ton $NCV_{p,kerosene} = 0.0438$ TJ/ton $NCV_{p,LPG} = 0.0473$ TJ/ton
Choice of data or Measurement methods and procedures	For Firewood and Charcoal, the methodology NCV defaults are applied. For Kerosene and LPG, the IPCC 2006 Vol2 Chap 1 Table 1.2 values are applied.
Purpose of data	To calculate parameters $SE_{p,y,CO2}$ and $SE_{p,y,non-CO2}$.
Additional comment	The methodology default emission factor has been applied since methodology default NCV is applied.

Data / Parameter	SDG 13 ICS 17: $f_{NRB,i,y}$
Unit	Percentage
Description	Fractional non-renewability status of woody biomass fuel during year y , in case the baseline fuel is biomass or charcoal
Source of data	Determined by following the CDM TOOL 30, version 3, Calculation of the fraction of non-renewable biomass
Value(s) applied	0.9447
Measurement methods and procedures	Calculated
Monitoring frequency	One of two options, with the option defined and fixed at project design certification stage: 1. Determined ex-ante and fixed for a given crediting period or 2. Updated biennially or at each monitoring and verification PP has chosen option 1 and the parameter has been fixed ex-ante for the crediting period
QA/QC procedures	Requirements of the CDM TOOL30
Purpose of data	Calculation of emission reductions
Additional comment	Updated values from the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Table 4.9 ³³ have been applied. The NRB shall be reassessed during renewal of the crediting period on most recent information available.

³³ (https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf)

Data / Parameter	SDG 13 ICS 18: $P_{b,y}$
Unit	t/household-day
Description	Quantity of fuel that is consumed in baseline scenario b during year y
Source of data	Baseline performance field tests
Value(s) applied	$P_{b,firewood} = 0.0069\text{t/day/household}$ $P_{b,charcoal} = 0.0017\text{t/day/household}$ $P_{b,kerosene} = 0.00062\text{t/day/household}$ $P_{b,LPG} = 0\text{ t/day/household}$
Measurement methods and procedures	The solid fuels (firewood and charcoal) were weighed using calibrated digital scales while kerosene was measured using a 1 litre calibrated measuring cup.
Purpose of data	Used to calculate parameter SE under method 3.
Additional comment	The values applied have been determined from the baseline KPTs survey results. The firewood values are within the threshold value of 0.75 tonnes/person*year of fuelwood. For kerosene, the measurement had been in liters, and this has been multiplied by 0.817 to convert to kgs (1 liter of kerosene in kg (coolconversion.com))

B.6.3. Ex ante estimation of SDG Impact

Ex ante calculations related to the outcomes of SDG 13

The transparent ex-ante calculations of the outcomes of SDG 13 (i.e. CO₂e reductions) are provided in a separate Excel Spreadsheet uploaded to SustainCert App. For data/parameters available before design certification values contained in section B.6.2 above and for data/parameters not available before design certification the estimates contained in section B.7.1 below have been used.

Ex ante calculations related to the outcomes of SDG 3

Proportion of users that perceive a reduction in smoke levels (%)

The VPA entails distribution and installation of metered LPG cookstove technology to replace the inefficient baseline cookstoves. The baseline stove emits smoke and other particulates which have implications on human health, while the project promotes use of LPG which is a clean fuel. Using the project technology is expected to lead to significant reduction in smoke levels among end-users. 94.6% of the baseline

survey respondents expressed they experienced high to moderate smoke levels by using the baseline technologies.

The ex-ante estimate for SDG 3 has been estimated based on the projected indoor improvement from using the project technology. The project is expected to lead to 100% improvement in indoor air quality immediately end users switch to the project fuel and stop using the baseline fuels.

Ex ante calculations related to the outcomes of SDG 5

Average time saving associated with cooking time and fuel collection (hours)

The use of the project technology will enable end-users helps to reduce the time spent on fuel collection and cooking which is mostly done by women. Fuel collection takes more time and effort, which takes away productive hours of women and young girls leading to time poverty and lost opportunity. As detailed on table 4 of the study report "Modelling Approaches and Performance for Estimating Personal Exposure to Household Air Pollution: A case Study in Kenya" done in Kenya by Johnson et.al and the Berkeley Air Monitoring Group³⁴, Household's cooking using LPG were found to spend a cooking time of 126 minutes (2.1hrs) compared to 330 minutes (5.5hrs) and 320 minutes (5.3hrs) spent by charcoal and firewood users respectively.

The ex-ante estimation for the parameter has been determined by calculating the difference between the time spent using LPG compared to baseline fuel, to obtain a value of 3.2 hours per day as shown below.

Time saving	Value applied
Average daily time spent in cooking using the baseline fuel	5.3 (value for charcoal since its more conservative)
Average daily time spent in in cooking using project fuel	2.1
Time savings (hrs)	3.2

³⁴ [Modeling approaches and performance for estimating personal exposure to household air pollution: A case study in Kenya - PubMed \(nih.gov\)](#)

Ex ante calculations related to the outcomes of SDG 7

Proportion of population with primary reliance on clean fuels and technology (number)

This ex-ante estimation for the parameter has been derived from the projected stove sales provided by the project developer. The project is expected to distribute about 1,928,113 LPG units among end-users.

Ex ante calculations related to the outcomes of SDG 8

Total number of employees earning above local minimum wage (number of people)

The VPA is expected to create job opportunities in its different departments. This ex-ante estimation for this parameter has been determined from the estimate provided by the project implementer. Each year the project is expected to employ about 1000 people who will include both males and females, who will be paid above the set national minimum wage for the host country.

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 13: Amount of GHGs emissions avoided or sequestered (tCO2e)

Year	Baseline estimate	Project estimate	Net benefit
02/09/2020 to 31/12/2020	5854.21	101.65	4,314.00
2021	391550.49	6798.73	288,563.25
2022	1279702.23	22220.24	943,110.75
2023	3327225.81	57772.62	2,452,089.75
2024	4934821.02	85686.27	3,636,850.50
01/01/2025 to 01/09/2025	3289880.68	57124.18	2,424,567.00
Total	13,229,034.00	229,703.00	9,749,495.00
Total number of crediting years		5	
Annual average over the crediting period	2,645,806.00	45,940.00	1,949,899.00

**SDG 8: Total number of employees earning above local minimum wage
(number)**

Year	Baseline estimate	Project estimate	Net benefit
02/09/2020 to 31/12/2020	0	333	333
2021	0	1000	1000
2022	0	1000	1000
2023	0	1000	1000
2024	0	1000	1000
01/01/2025 to 01/09/2025	0	667	667
Total	0	5000	5000
Total number of crediting years	5		
Annual average over the crediting period	0	1000	1000

Source: GS11331 SDG Impact tool

SDG 7: Proportion of population with primary reliance on clean fuels and technology (number)

Year	Baseline estimate	Project estimate	Net benefit
02/09/2020 to 31/12/2020	0	6862	6862
2021	0	152985	152985
2022	0	347014	347014
2023	0	800000	800000
2024	0	310626	310626
01/01/2025 to 01/09/2025	0	310626	310626
Total	0	1928113	1928113
Total number of crediting years	5		
Annual average over the crediting period	0	385,623	385,623

Source: GS11331 SDG Impact tool

SDG 5: Average time saving associated with cooking time and fuel collection (hours)

Year	Baseline estimate	Project estimate	Net benefit
02/09/2020 to 31/12/2020	2.65	1.05	1.6
2021	5.3	2.1	3.2
2022	5.3	2.1	3.2
2023	5.3	2.1	3.2
2024	5.3	2.1	3.2
01/01/2025 to 01/09/2025	2.65	1.05	1.6
Total	26.5	10.5	16
Total number of crediting years	5		
Annual average over the crediting period	5.3	2.1	3.2

Source: GS11331 SDG Impact tool

SDG 3: Proportion of users that perceive a reduction in smoke levels (%)

Year	Baseline estimate	Project estimate	Net benefit
02/09/2020 to 31/12/2020	0%	100%	100%
2021	0%	100%	100%
2022	0%	100%	100%
2023	0%	100%	100%
2024	0%	100%	100%
01/01/2025 to 01/09/2025	0%	100%	100%
Total	0%	100%	100%
Total number of crediting years	5		
Annual average over the crediting period	0%	100%	100%

Source: GS11331 SDG Impact tool

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

Data / Parameter	SDG 13 ICS 15: Avoidance of double counting or double claiming among project technology end users
Unit	NA
Description	Evidence of avoidance of double counting or double claiming with project technology end users
Source of data	Evidence of informing / notification of end users, such as: - Leaflets distributed with the warranty card of the product alerting end-users to the waiving of their carbon rights in exchange for discount pricing of the improved technology below its true cost, - Carbon title waiver forms signed by end users, etc.
Value(s) applied	Evidence of carbon waiver transfer to be provided during verification.
Measurement methods and procedures	N/A
Monitoring frequency	To be monitored whenever a project technology is sold or otherwise disseminated
QA/QC procedures	PP shall cross check using general internet search and search of public records of Gold Standard and other voluntary market and UNFCCC mechanisms
Purpose of data	To demonstrate avoidance of double counting or double claiming among project technology end users
Additional comment	The PP shall provide evidence of informing / notification of end users on the transfer of carbon rights.

Data / Parameter	SDG 13 ICS 16: Presence of stove stacking
Unit	NA
Description	Descriptive statistics of the presence and usage practices of baseline- and other non-project-technology by project technology end users.
Source of data	The methodology provides that any one of the following methods may be applied: - Measurement campaigns shall be undertaken using data loggers such as stove utilization monitors (SUMs) which can log the operation of all devices in order to determine the average device utilization intensity, or - Usage Survey- use of other stoves, to capture cooking habits and stove usage of households in the region, including quantification of use of baseline devices, by formulating questions and/or collecting evidence to determine the frequency of usage of both the project devices and baseline devices, or monitoring surveys to capture the number of meals cooked. The surveys may be integrated with the usage survey
Value(s) applied	0
Measurement methods and procedures	Measured
Monitoring frequency	Annual
QA/QC procedures	The calculation of SE_{b,y,CO_2} and/or $SE_{b,y,non-CO_2}$ shall be cross-checked with the observed presence of stove stacking. PP will ensure any stove stacking is considered so that emission reductions are calculated only from real reduction of, or replacement of, baseline fuel use.

	PP will also cross-check results of this survey with independent studies that are specific to the project region (or to the project country, if regional studies are not available), including but not limited to National publications, peer-reviewed literature, third party assessments (for example – WISDOM, FAO, UN and similar organizations) and/or official data or statistics about cooking technologies, not older than 5 years old.
Purpose of data	Calculation of emission reductions
Additional comment	As required under the methodology, whether or not the existing baseline technology is surrendered, when an old technology remains in use in parallel with the improved technology, or another technology is put in use in parallel, the corresponding emissions shall be accounted for so that emission reductions are not overestimated. The PP assumes that there will be no presence of stove stacking in the project scenario and a value of 0 is applied for ex-ante estimation purposes.

Data / Parameter	SDG 13 ICS 19: $P_{p,y}$
Unit	kg/household-day
Description	Quantity of fuel that is consumed in project scenario p during year y. LPG, Kerosene, Firewood and Charcoal fuel consumption will be monitored.
Source of data	Project performance field tests
Value(s) applied	$P_{p,firewood} = 0$ t/day/household $P_{p,charcoal} = 0$ t/day/household $P_{p,kerosene} = 0$ t/day/household $P_{p,LPG} = 0.000136$ t/day/household
Measurement methods and procedures	Calculated
Monitoring frequency	Updated every two years, or more frequently. The KPT values shall remain valid for two years and may be applied for before or after period, however the gap between start date of first KPTs and second KPTs shall not be more than two years.
QA/QC procedures	Compliance with the general requirements for sampling, general requirements for QA/QC and Kitchen performance test.
Purpose of data	Used to calculate SE under method 3.
Additional comment	The Circle Gas Smart meter technology allows for real time measurement of LPG consumption per household. The value for the parameter will be obtained by calculating the average consumption for each end-user from the daily usage data maintained by the project for each end-user. The value of 136g/day applied is a 7-day average obtained from the registered end-users as per the Circle Gas customers fuel consumption database by the time of submission to the VVB for validation. Actual value to be determined during monitoring.

Data / Parameter	SDG 13 ICS 22: SE_{b,y,CO_2}
Unit	tCO ₂ /technology*day
Description	Specific CO ₂ emissions for a baseline b technology in year y
Source of data	Calculated from $P_{b,y}$, EF_{b,f,CO_2} and other information to obtain the emissions of the required units
Value(s) applied	0.00629
Measurement methods and procedures	Calculated
Monitoring frequency	Updated every two years, or more frequently
QA/QC procedures	The calculation method and inputs has been described in detail in the VPA-DD and will be included in the monitoring report.
Purpose of data	Calculation of emission reductions
Additional comment	The value is a weighted average based on baseline fuel consumption using the data obtained from the baseline KT _s . The value is obtained by multiplying $P_{b,y} * EF_{p,CO_2} * NCV_b$

Data / Parameter	SDG 13 ICS 23: SE_{p,y,CO_2}
Unit	tCO2/technology*day
Description	Specific CO2 emissions for a project p technology in year y
Source of data	Calculated from $P_{p,y}$, EF_{p,f,CO_2} and other information to obtain the emissions in the required units
Value(s) applied	0.000101
Measurement methods and procedures	Calculated
Monitoring frequency	Updated every two years, or more frequently
QA/QC procedures	The calculation method and inputs has been described in detail in the VPA-DD and will be included in the monitoring report.
Purpose of data	Calculation of emission reductions
Additional comment	The value is obtained by multiplying P_{py} * $EF_{p,CO_2} * NCV_p$

Data / Parameter	SDG 13 ICS 24: $SE_{b,y,non-CO2}$
Unit	tCO ₂ /technology*day
Description	Specific non-CO ₂ emissions for a baseline b technology in year y
Source of data	Calculated from $P_{b,y}$, $EF_{b,f,nonCO2}$ and other information to obtain the emissions in the required units
Value(s) applied	0.00107
Measurement methods and procedures	Calculated
Monitoring frequency	Updated every two years, or more frequently
QA/QC procedures	The calculation method and inputs has been described in detail in the VPA-DD and will be included in the monitoring report.
Purpose of data	Calculation of emission reductions
Additional comment	The value is a weighted average based on calculation using the fuel consumption data obtained from the baseline KT _s . $P_{by} * EF_{b,non-CO2} * NCV_b$.

Data / Parameter	SDG 13 ICS 25: $SE_{p,y,non-CO2}$
Unit	tCO2/technology*day
Description	Specific non-CO2 emissions for a project p technology in year y
Source of data	Calculated from $P_{p,y}$, $EF_{p,f,nonCO2}$ and other information to obtain the savings in the required units
Value(s) applied	0.000026
Measurement methods and procedures	calculated
Monitoring frequency	Updated every two years, or more frequently
QA/QC procedures	The calculation method and inputs has been described in detail in the VPA-DD and will be included in the monitoring report.
Purpose of data	Calculation of emission reductions
Additional comment	The value is obtained by multiplying $P_{py} * EF_{p,non-CO2} * NCV_p$

Data / Parameter	SDG 13 ICS 26: $U_{p,y}$
Unit	Percentage
Description	Weighted average usage rate in project scenario p during year y
Source of data	Usage survey
Value(s) applied	75%
Measurement methods and procedures	Calculated
Monitoring frequency	At least annually or more frequently
QA/QC procedures	Compliance with the general requirements for sampling and general requirements for QA/QC.
Purpose of data	Calculation of emission reductions
Additional comment	The value applied is an estimate for purposes of ex-ante calculation, actual value to be determined during monitoring.

Data / Parameter	SDG 13 ICS 27: $N_{b,p,y}$
Unit	Days
Description	Number of project technology-days included in the project database for baseline b/project p pair in year y
Source of data	Calculated from the Project database as the sum of the number of project technology units' times the calendar days during the year y that they were present at the end user locations
Value(s) applied	365
Measurement methods and procedures	Calculated
Monitoring frequency	Annually
QA/QC procedures	PP shall cross check the results of the usage survey with the contents of the project database to confirm whether the project technology units surveyed are present at end user locations as expected, or not. If there is discrepancy, this shall be explained or corrected.
Purpose of data	Calculation of emission reductions
Additional comment	The value applied is an estimate for purposes of ex-ante calculation, actual value to be determined during monitoring.

Data / Parameter	SDG 13 ICS28: LE_{p,y}
Unit	tCO ₂ e per year
Description	Leakage in project scenario p during year y
Source of data	Sources established by following Leakage emissions assessment requirements
Value(s) applied	0
Measurement methods and procedures	Calculated
Monitoring frequency	Every two years, or default discount value of 0.95 applied to emission reductions
QA/QC procedures	Compliance with the general requirements for sampling and general requirements for QA/QC
Purpose of data	Calculation of emission reductions
Additional comment	An assessment of leakage has been demonstrated in section B.6.1 of the VPA-DD. From the assessment, there is no perceived leakage from the project activity.

SDGs Monitoring

Data / Parameter	SDG 3 Proportion of users that perceive a reduction in smoke levels.
Unit	Percentage
Description	Proportion of end-users using the project technology who experience an improvement in indoor smoke levels.
Source of data	Project survey
Value(s) applied	100%
Measurement methods and procedures	Questions will be included in the survey questionnaire to assess the parameter. Survey results will determine the proportion of population who demonstrate achieving an improvement in indoor smoke level.
Monitoring frequency	Annual
QA/QC procedures	The outcomes of the survey to be compared with the baseline scenario results.
Purpose of data	Reporting on SDG impacts
Additional comment	67.7% and 26.9% of the baseline survey respondents reported experiencing high and moderate indoor smoke levels respectively due to the use of the baseline fuels and stove technologies. The level of indoor smoke in end-user's premises is expected to reduce significantly as a result of using the project technology. The value applied is an estimate for purposes of ex-ante calculation, actual value to be determined during monitoring.

Data / Parameter	SDG 5 Average time saving associated with cooking time and fuel collection
Unit	hours/household/day
Description	Average time saving associated with cooking time and fuel collection in the project scenario.
Source of data	Project records and surveys
Value(s) applied	3.2
Measurement methods and procedures	Questions will be included in the survey questionnaire to assess the parameter. Survey results will determine the average time saving associated with cooking time and fuel collection among end-users.
Monitoring frequency	Annual
QA/QC procedures	-
Purpose of data	Reporting on SDG impacts
Additional comment	As detailed on table 4 of the study report "Modelling Approaches and Performance for Estimating Personal Exposure to Household Air Pollution: A case Study in Kenya" done in Kenya by Johnson et.al and the Berkeley Air Monitoring Group, Household's cooking using LPG were found to spend a cooking time of 126 minutes compared to 330 (5.5hrs) and 320 (5.3hrs) minutes spent by charcoal and firewood users respectively. The estimated time saving is given as the difference of time spent using LPG compared to baseline fuel. The time spent in cooking using the project technology is expected to reduce compared to the baseline scenario due to improved energy efficiency.

Data / Parameter	SDG 7 Proportion of population with primary reliance on clean fuels and technology
Unit	Number
Description	The number of people making use of the project technology.
Source of data	Project distribution database
Value(s) applied	1,928,113
Measurement methods and procedures	The distribution records will be checked to determine the number of people using the project technology.
Monitoring frequency	Annual
QA/QC procedures	An updated distribution database shall be maintained by the project developer.
Purpose of data	Reporting on SDG impacts
Additional comment	The value applied is the projected number for the first crediting period. Actual number will be determined during monitoring.

Data / Parameter	SDG 8 Total number of employees earning above local minimum wage
Unit	Number
Description	The number of people employed by the project who are earning above local minimum wage for the host country.
Source of data	Employment records
Value(s) applied	1,000
Measurement methods and procedures	The number of people employed, and their income levels will be provided by the project developer. This will be crosschecked against the minimum wage for the host country.
Monitoring frequency	Annual
QA/QC procedures	-
Purpose of data	Reporting on SDG impacts
Additional comment	The value applied is the projected number. Actual number will be determined during monitoring.

B.7.2. Sampling plan

a) General requirements for sampling

Section 4.4. of the applicable methodology outlines the general requirements for sampling and provides that when sampling is applied to determine mean (average) parameter values or proportion (yes/no) parameter values for both ex-ante and monitored data and parameters, the outlined guidelines shall always be applied. Furthermore, specific requirements apply for the sampling related to some parameters.

The sampling method chosen for the project activity is Simple Random Sampling. This will be done in accordance with the Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 4.

The project developer shall maintain an updated distribution record for all the units, and this will identify the end-users and the geographical zones of operational project technologies within the project activity. Random test subjects shall hence be identified from the distribution.

Target population

The target population will be the total number of end-users using the project technology.

Sampling Methodology

One unit of baseline cookstove in each end-user premises will be replaced with one metered LPG cooking unit. Since the cooking patterns and behaviours are expected to be the same across the VPA boundary, the usage pattern will most likely be similar. The project will employ simple random sampling method across the VPA when drawing up a sample. Simple random sampling is suited to populations that are homogeneous.

Sample Size

In accordance with general requirements for sampling stipulated in Section 4.4 of the methodology, the PP shall determine sample sizes using the applicable precision level for the different parameters to be monitored by the project.

A statistically valid sample can be used to determine parameter values, as per the relevant requirements for sampling in the "Methodology for Sampling and surveys for CDM project activities and programme of activities." Minimum 90% confidence interval and a 10% margin of error requirement shall be achieved for the sampled parameters.

In any case, for proportion parameter values, a minimum sample size of 30, or the whole group size if this is lower than 30, must always be applied.

When a baseline and project survey is used the following sample size guidelines should be applied, unless otherwise stated for specific parameters:

- Group size <300: Minimum sample size 30 or population size, whichever is smaller.
- Group size 300 to 1000: Minimum sample size 10% of group size

- Group size > 1000 Minimum sample size 100

For usage survey, the value of interest is a proportion parameter, the following equation for the sample size will be used to calculate sample size when applying simple random sampling (As provided in the *Guidelines for Sampling and Surveys for CDM Project Activities and Programme of Activities'*, Version 04.0³⁵).

$$n \geq \frac{1.96^2 NV}{(N-1) \times 0.1^2 + 1.96^2 V} \quad \text{Equation (40)}$$

Where: $V = p(1-p)/p^2$

N = Sample size with finite population correction

P = Estimated proportion

1.96 = Represents the 95% confidence required

0.1 = Required precision

General Requirements for Quality Assurance and Quality Control

- The project developer is responsible for accurate and transparent record keeping, monitoring and evaluation. All supporting documentation and records for the project must be easily accessible for spot checking and cross referencing by a third-party auditor.
- Contact information in the total distribution record shall allow a project auditor to easily contact and visit end users.
- An auditor must be able to cross reference pertinent project documentation, including archives such as production records (e.g., materials purchases, internal logs), financial accounts and sales records, as well as wholesale customer invoices, observations of retailer activities and sales performance.

³⁵ [Meth_GC48 \(ver04.0\) \(unfccc.int\)](#)

B.7.3. Other elements of monitoring plan

The following shall be monitored and recorded during project crediting period.

a) Total sales or dissemination record

The project developer shall maintain an accurate and complete “dissemination record” or “installation record.” The record shall be maintained continuously and backed up electronically. The required data for each project unit is:

1. Date of installation
2. Geographic area of sale
3. Model/type of project technology sold
4. Quantity of project technologies sold
5. Name, telephone number (if available), and address and/or GPS coordinates: a. Required for all bulk purchasers, i.e., retailers and industrial users; b. Required for all end users, e.g. households
6. Mode of use: domestic, institutional commercial, other.

b) Project Database

The project database lists all the project technology units that have been distributed by the project and have not surpassed their technical life. It will be derived from the total dissemination record and shall be maintained continuously. Within the database, project technologies units are labelled, at a minimum, with their corresponding project scenario p and their date of dissemination. Technologies aged beyond their technical life, and not replaced or retrofitted, are removed from the project database, and no longer credited.

c) Project Surveys

The project surveys include the usage survey ($U_{p,y}$), Usage Survey- use of other stoves (ICS 16), and the baseline and project performance field tests (BFT and PFT).. The project surveys have the same sample sizing and data collection guidelines as the baseline scenario survey. The project surveys shall only be conducted with end user’s representative of the project scenario and currently using the project technology.

j) Usage Survey ($U_{p,y}$)

The objective of the usage survey is to provide a single usage parameter that is weighted based on the age distribution for project technologies in the Project

Database, Up,y . This usage parameter must be established to account for drop off rates as project technologies age and are replaced, or for reduced use by end users for other motives.

The usage survey shall determine the usage proportion for each project technology being credited for each project scenario p .

The usage survey shall be conducted in line with the requirements and guidelines of sampling.

ii) Usage Survey- use of other stoves

As part of the usage survey, the project developer shall collect data on the presence and usage practices of baseline and other non-project technology by project technology end users and prepare descriptive statistics of these practices. The same method of in person interviews and expert observation within the kitchen in question will be used to collect these data.

iii) Field Tests

The baseline and project performance field tests (BFT and PFT) measure real, observed technology performance in the field. Since the project applies method 3, a field test was carried out for baseline scenarios by testing an independent sample. In the project scenario, this will be determined based on actual data since the project uses a metered technology to monitor fuel consumption for each end-user.

The project developer shall design the field test to ensure monitoring is representative of technology and fuel use practices. The end-users will be required to behave and consume fuel normally, using the technologies they normally use. Participants will not be influenced to use a specific stove or fuel during the study, nor deviate from typical stove and fuel practices.

d) Baseline Fuel Consumption

To calculate SE under method 3 applied, the project developer has determined Pb,y , *Quantity of fuel that is consumed in baseline scenario b during year y in units of kg/household-day, kg/person-meal, etc.* The baseline fuel consumption was determined using a project specific Baseline Performance Field Tests (BFT). The BFT has been completed once ex-ante, and will be revised during crediting period renewal. The following requirements were applied for selecting the end user locations of the Baseline Performance Field Tests:

- The BFT reflected the average household size within the project area.

- The BFT took place in those households where improved devices had not yet been disseminated and where baseline technology was still in use in the project area.

e) Project Fuel Consumption

To calculate SE under method 3 applied, the project developer shall determine $P_{p,y}$, *Quantity of fuel that is consumed in project scenario p during year y in units of kg/household-day, kg/person-meal, etc.* The project fuel consumption shall be determined using a project specific Project Performance Field Test (PFT). The PFT is completed every other year, or more frequently, and the first PFT must be completed before the first verification of the project. The PP shall use actual fuel consumption data maintained for each end-user since the project makes use of a metered technology that allows for real-time fuel consumption monitoring. The following requirements shall apply for selecting the end user locations of the Project Performance Field Tests:

- End users shall be representative of the project scenario target population and currently using the project technology.
- The PFT is a study of the fuel consumption of the cooking practices of project technology end users during the crediting period. The PFT shall evaluate whether the project technology provides the same services at the same/similar frequency as the baseline technology. As such, when the project technology does not replace or reduce the entire baseline fuel consumption, the calculation SE shall adjust the baseline fuel consumption downward based on the cooking services replaced by the project stove.
- Where the baseline technology or other non-project technologies operate as backups or complementary units in parallel with project technology ("stove stacking"), the project fuel consumption implications shall be accounted for in the PFT.

SECTION C. DURATION AND CREDITING PERIOD

Duration of project

C.1.1. Start date of project

02/09/2020 (date of first installation of the LPG units to end-users)

C.1.2. Expected operational lifetime of project

20 years

Crediting period of project

C.1.3. Start date of crediting period

02/09/2020 (start date of project operation).

C.1.4. Total length of crediting period

5 years (renewable twice)

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarized below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 2	Project developer shall monitor the number of people employed by the project including their gender and income to ensure equal opportunities are given to both men and women, and that equal pay for equal value of work done is maintained.
Principle 3	The PP shall check and monitor occupational health and safety measures that have been established at the LPG depot including fire preparedness.
Principle 6.1	Work agreements/contracts will be signed by the permanent staff employed by the project detailing the working agreements as required.
Principle 6.1	Employment records shall document the ages of the persons employed by the project. These records shall be provided as supporting evidence to confirm that no children are employed by the project.
Principle 6.1	The PP shall monitor incidences of fire outbreaks incurred by end-users using the project technology. The safety information shared with end-users will also be provided as supporting evidence.
Principle 9.5 Hazardous and Non-hazardous Waste	The project implementor shall ensure that all used batteries are collected and keep an annual record of the number of batteries collected and taken for recycling to approved recycling site.

D.2 Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project developer seeks to respect gender sensitivity in the design and implementation of the project. Any gender inequalities and gender-related risks that may be identified in project gender analysis shall be adequately addressed. Further, a gender-sensitive approach will be adopted during stakeholder consultation by promoting equitable information sharing between women and men stakeholders, which will be made available and presented in an accessible format across all the stakeholder groups including those more marginalized. The approach will also include opportunities for stakeholders to share information in a two-way exchange, give regular feedback during implementation and ensure their views and priorities are incorporated in project design and implementation. The project has further provided quantifiable 'easy-to-measure' indicators potentially aligned to the related SDG goals that measure gender-related gaps and risks for the project. Efforts shall also be geared towards innovatively increasing women's participation in the project since they are the key beneficiaries of the project.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Government of Kenya adopted the National Policy on Gender and

	Development in 2000. The policy spells out a policy approach of Gender mainstreaming and empowerment of women and clearly states that it is the right of women, men, girls and boys to participate in and benefit equally from the development process ³⁶ . The plan of action (2008 – 2012) to implement the national policy on gender and development with an objective of facilitating the mainstreaming of gender concerns in all areas of the development process and to provide a basis for stakeholders to initiate programmes to promote gender equality in the country ³⁷ . The project will provide job opportunities during its overall operation which will benefit both men and women.	
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No	
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No	

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

³⁶ <http://www1.uneca.org/Portals/ngm/Documents/GenderPolicy.pdf>

³⁷ http://www1.uneca.org/Portals/ngm/Documents/Gender_Mainstreaming%20Action%20Plan%20final%20-Apr08.pdf

E.1. Summary of stakeholder mitigation measures

A local stakeholder consultation meeting was conducted on 05/08/2021 at the MovenPick Hotel, Westlands, Nairobi Kenya. The detailed stakeholder consultation report has been submitted to the Gold Standard. Below is a summary of the comments that were raised during the consultation.

Gender	Comment	Was the comment taken into consideration	How? Why?
M	How is M-Gas working with the community to reduce fire incidences? Do you have someone in place to ensure safety?	Yes	The project only uses 13kilogram cylinder which cannot burst in case of a fire since it comes with a release valve which prevents it from bursting. Additionally, before installation of the gas, the household is taken through training on best practices to use it, including the M-Gas staff positioning the gas in a place away from things that can easily catch fire like clothes. Besides, M-Gas works closely with youth leaders, chiefs and employ staff who come from the same area as where they are installing the project for ease of tracing in case of a fire incident
M	In case I lose my smart card, can it be replaced? What security measures do you have in place to ensure there is no theft of the smart meter? Do you replace the smart meter in	Yes	First time you lose the smart card, replacement is free, however 2 nd time, you pay a fee of

	case of damage? In case fire starts in my house, how do I stop it?		Ksh. 100. As for security, we are able to monitor the movement of the smart meter remotely hence in case there is theft, we can retrieve it. In case a stakeholder wants to move to a new location, they cannot move with the gas, they would have to get a new set in the new area they are moving so that the smart meter is registered with the new location. Smart meters are specific to specific areas for ease in follow-up. In case a fire starts, the trained personnel are a phone call away, you can reach them. Also, the fire marshals can help. Additionally, remember that the gas stops running once the money deposited is depleted. That way, damage is minimised.
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M	Are you involving youths during the implementation of the project? When are you going to bring this project in our area (Kangemi)?	Yes	Yes, we mostly work with youths by employing them and taking them through training. We are working on bringing the project to your area. At the moment M-Gas is working on getting the government approval to operate in Kangemi.
M	Can you give a rough estimate on the relationship between amount of money loaded on the smart meter, versus time it will last to cook? When are you coming to our area (Dandora)?	Yes	Roughly, Kshs. 10 can make tea, Kshs. 10-15 can make Ugali for a household of about 4-5 people. We are hoping to come to Dandora soon once the logistics are put in place.
F	Can you install M-Gas to a customer who is already using LPG? Are your cylinders tested by Kenya Bureau of standards?	Yes	The primary focus of the project is to provide clean energy to those currently using 'dirty fuels' so no. Yes, the gas cylinders are tested by Kenya Bureau of Standards annually.
F	If I have an old gas cylinder, can I exchange it for the M-gas cylinder?	Yes	No, as mentioned, primary focus is the end-users currently using 'dirty fuel', also

			we prefer to work with our own cylinders as we are sure of the quality of products used.
M	What is the warranty of the products?	Yes.	M-Gas will re-validate the gas cylinders after 7 years of use. For the stoves being used, they are yet to come-up with a timeline as they have only been used for 18 months. Once they are able to gather enough data, they shall then be able to come up with re-validation timelines.
M	How can I reduce exposure to risk of people coming to my house pretending to be M-Gas staff, yet they are just thief's? Is there any form of identification?	Yes.	Before coming to each house, the client is called by M-Gas to notify them that they are on the way. Further M-Gas has branded T-shirts and Identification tags to help houses are able to correctly identify them.
F	How do you ensure safety of M-Gas staff visiting houses?	Yes.	M-Gas walk in pairs who operate in proximity of each other. Additionally, the phones the staff carry have a

			panic button on them.
F	Do I get a warning when my gas is almost depleted on the cylinder?	Yes.	Smart meter helps one see the levels of the gas hence one can tell when it's almost depleted one can contact customer service representatives and ask for replacements. Also, M-Gas staff are able to monitor from their end and do replacements.
M	Why are there some incidences of payments taking too long to reflect after paying?	Yes	This could be a network coverage issue in some areas; however, M-Gas is working very hard to constantly do upgrades to ensure they are giving the best support to customers and avoid such delays
M	Can M-Gas work with locals such as <i>Nyumba Kumi</i> to help show the way?	Yes	M-Gas employs local people from an area to ensure that tracing households isn't much of a challenge.
F	Do you employ staff with disability?	Yes	Yes, there is no discrimination. Mr. Yugi noted that they have a special

			employee who serves end-users with sight and hearing problems.
M	Do you currently have initiatives to support the community?	Yes	At the moment, the main focus is offering the gas cylinder for free which comes with the cooking stoves to promote clean cooking. In future, M-Gas will consider additional activities it can include as Corporate Social Responsibility (CSR).
M	How will you prevent "false" accounting of emission reductions from households switching from the convectional LPG cylinders to the ones proposed?	Yes.	We are not going to claim emission reductions for households switching from convectional LPG to the project LPG. The project is targeting households using non-renewable biomass and kerosene.
M	What is the baseline scenario? Is it charcoal use, wood use? How real is the baseline?	Yes	Our baseline scenario is a mix of firewood, charcoal and kerosene use. This has been informed by preliminary studies that we have done

			(mostly in Nairobi slum areas where installations have been done) which indicate that most households apply a mix of the three fuels.
M	What economic incentive(s) does the project offer to the consumer so as to be attractive compared to similar alternatives? Cost, convenience? The economic incentive should be over and above the quality objectives.	Yes	The project is offering the complete LPG unit (stove, LPG cylinder fitted with a smart meter and the pipe) for free to the end-users as an incentive. This will remove the initial capital required to procure the conventional LPG units and the end-user will only be required to buy the fuel. Further, the project is adopting the Pay as You Go approach to allow end-users purchase fuel through Mpesa based on available money and they don't have to do a full refill. The smart meter also allows M-Gas to track fuel usage and refill can be scheduled before the

			household runs out of fuel, and delivery is free. These provisions are aimed at accelerating access to LPG in a reliable, convenient, and affordable way, and at the same time ensure households do not revert to using the baseline fuels.
M	What are the safety measures in place to prevent leaks in the metering system considering it is preinstalled? For the conventional cylinders, there is a cork/plug that keeps the cylinder shut until a regulator is installed	Yes	The smart meter allows for leak detection. In case of a leakage, M-Gas will be able to detect this from their end and will immediately disconnect the supply and inform the household. M-Gas will then send a representative to check and service the unit. As a safety measure, the installation, refills, and servicing of the units is done by the M-Gas sales and service teams who are highly trained.

Stakeholder Feedback Round (SFR)

The stakeholder feedback round (SFR) commenced on 24/11/2021 and covered a period of 60 days till 24/01/2022. The consultation was conducted online with an aim of getting further feedback from the stakeholder on the PoA design and implementation of the project. The consultation was done with all the stakeholders who attended the meeting and those who were invited though they did not attend. The documents shared with the stakeholders included:

- The local stakeholder consultation report
- Project of Activities Design Document (PoA-DD)
- Voluntary Project Activity (VPA-DD)
- The Gold Standard Preliminary Review feedback form.
- The Gold Standard PoA Design Consultation Review final feedback form
- Non-technical summary in English
- Stakeholder feedback form

No comments were received from this consultation.

E.2. Final continuous input / grievance mechanism

Method	Method Chosen (include all known details e.g. location of book, phone, number, identity of mediator)
Continuous Input / Grievance Expression Book (mandatory)	M-Gas main depot in Nairobi and all other local depots
GS Contact (mandatory)	help@goldstandard.org
Telephone access (optional)	M-Gas Customer Service number 0800721148
Internet/email access (optional)	customercare@mgas.ke
Nominated Independent Mediator (optional)	See table below
Other	N/A

Nominated Independent Mediators

No.	Name	Location	Phone number
1.	Dominic Musembi	Embakasi	0746507316
2.	Ouma Osera	Mathare (Huruma)	0720636215
3.	Faith Wanjiru	Kangemi	0721744083
4.	Halima Hanii	Korogocho	0701369628
5.	Fiu Nifu	Karura Muthaiga	0710313552
6.	Lawrence Rapando	Kawangware	0728099312
7.	Moses Omondi	Kibera	0723176472
8.	Meima Ibrahim	Kibera	0722247987
9.	Njiru Mwangi	Lucky summer	
10	Francis Wambua	Kangemi	0702519337
11	Kennedy Ochieng	Dandora	0710915886
12.	Hesbon Mong'are	Dagorreti South	0791339081

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights	No	The project is not complicit in any form of human rights abuses. The project activity shall respect human rights during its operation and there shall be no form of violence or human right abuses.	No mitigation required.
2. The Project shall not discriminate with regards to participation and inclusion	No	The project activity shall adopt a non-discrimination approach to ensure equal inclusion and participation of all including men and women in the	No mitigation required.

		opportunities that arise from the project.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women.	No	The project does not directly or indirectly lead to gender-based discrimination, nor does it contribute to adverse impacts on gender equality. Women will particularly benefit from the project through increased fuel savings, reduced cooking time and improved health.	No mitigation required.
2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work.	Yes	In respect to the National Gender and Equality Commission Act, the project developer seeks to promote gender equality and the empowerment of women. The project shall not contribute to discrimination against women, nor does it reinforce gender-based discrimination and/or inequalities.	Project shall monitor the number of people employed by the project including their gender and income to ensure equal opportunities are given to both men and women, and that equal pay for equal value of work done is maintained.
3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in	No	The Government of Kenya adopted the National Policy on	No mitigation required.

assessing gender risks. 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	No	Gender and Development in 2000³⁸. The policy spells out a policy approach of Gender mainstreaming and empowerment of women and clearly states that it is the right of women, men, girls and boys to participate in and benefit equally from the development process. In compliance with this policy, the project activity shall seek to address any potential gender risks by ensuring equal participation of both men and women in the project. Expert stakeholder has not been included in the assessment.	No mitigation required
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid	Potentially	The project promotes the use	The PP shall check and

³⁸ [NATIONAL POLICY ON GENDER AND DEVELOPMENT](#)

community exposure to increased health risks and shall not adversely affect the health of the workers and the community		of LPG fitted with a smart meter among end-users. Using this technology shall lead to improved health because of reduced household air pollution associated with 'dirty' baseline fuels such as firewood and kerosene. As a result, the project shall contribute towards reduction of community exposure to increased health risks associated with indoor pollution. However, LPG is a highly flammable fuel which may pose a health risk if not properly stored. In line with the Occupational Safety and Health Act of 2007, the project implementor shall ensure adequate safety measures are put in place at the LPG depots to avoid any incidence of fire that may pose a safety risk to the workers or the community.	monitor occupational health and safety measures that have been established at the LPG depot including fire preparedness.
Principle 4.1 Sites of Cultural and Historical Heritage			

Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional, or religious values or intangible forms of culture?	No	The project does not involve and is not complicit in the alteration, damage or removal of any critical historic, cultural, traditional, or religious heritage. The project activity entails distribution and installation of LPG units among end-users.	No mitigation required
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project does not involve and is not complicit in involuntary resettlement, displacement, or eviction of locals. The project activity involves distribution and installation of metered LPG cooking technology among end-users and no form of relocation is required.	No mitigation required
>>			
Principle 4.3 Land Tenure and Other Rights			
a.Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	The proposed project activity involves distribution and installation of metered LPG cooking technology among targeted end-users. There	No mitigation required

b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?		shall be no change in land tenure arrangements, usage rights or land ownership in the course of implementation of the project activity.	
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	The project recognizes and respects the rights of indigenous peoples and shall seek to respect, protect, and conserve the cultural, intellectual, religious, and spiritual property of indigenous people. However, the proposed project activity involves distribution and installation of metered LPG cooking technology and shall have no negative impacts on indigenous people.	No mitigation required
>>			
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce	No	The project does not involve and is not complicit in any form of corruption. The project activity shall not	No mitigation required

corruption or corrupt Projects		engage in, contribute to, or reinforce corruption of any kind during its implementation and all activities will be done in a transparent manner.	
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions.	No	Yes, the project does not involve and is not complicit in any form of forced or compulsory labour. Kenya enacted the Employment Act in 2007 ³⁹ and the project developer shall ensure that employment shall be in compliance with the laid out national and international laws and procedures.	No mitigation required.
2. Workers shall be able to establish and join labour organisations.	No	The project developer has not	No mitigation required.

³⁹ [Microsoft Word - THE EMPLOYMENT ACT 2007 \(labourmarket.go.ke\)](#)

3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not	Yes	set any restrictions on establishment and joining labour organizations. This will be at the discretion of the employees. The project developer shall ensure proper documentation for all employees through work agreements for the permanent employees.	Work agreements/contracts will be signed by the permanent staff employed by the project detailing the working agreements as required.
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including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion)	Yes	The project shall ensure that no children (below 18years of age) are employed by the project.	Employment records shall document the ages of the persons employed by the project. These shall be provided as supporting evidence to ascertain no children are employed by the project.
5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures	Potentially	The project implementor shall conduct relevant trainings to the field sale and service teams. Installation of the metered LPG cooking technology is a simple procedure that does not require sophisticated equipment. To ensure safety is maximised during the use of the units, the project implementer will sensitise end-users on the proper use of the LPG unit. The project smart meter is	The PP shall monitor incidences of fire outbreaks incurred in households from using the project technology. The safety information shared with end-users will also be provided as supporting evidence.

		designed to detect leakage and the gas is automatically shut off in case there is a leak hence maximising on safety.	
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	The project will not cause negative economic consequences but will rather create jobs for people employed as field sales and service teams, and overall management of the project which will have positive economic consequence.	No mitigation required
>>			
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The project entail use of LPG which will enable a shift from using baseline fuels such as non-renewable biomass (fuelwood and charcoal) and kerosene, that emit greenhouse gases. This will lead to reduced GHG emissions.	No mitigation required
>>			
Principle 7.2 Energy Supply			
Will the Project use energy	No	The project activity is an LPG	No mitigation required

from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?		project and does not involve use of energy supply from a local grid or fuel resource that provides for other users.	
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project activity entails distribution and installation of complete metered LPG cooking technology in end-user premises and does not entail use of water. The project will therefore have no impact on natural water patterns or flow.	No mitigation required
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion? b. Is the Project's area of influence susceptible to excessive erosion and/or	No	The project activity will promote a shift from use on non-renewable biomass to LPG, leading to reduced deforestation in the project boundary. This will consequently lead to reduction in the occurrence and severity	No mitigation required

water body instability?		of soil erosion.	
>>		The project area is not susceptible to excessive erosion or water body instability and the proposed project activity shall not negatively affect soil conditions in the project scenario.	
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project is an LPG project and does not involve use of land and soil for the production of crops.	No mitigation required
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The project does not create nor increase vulnerability to natural disasters. Using the metered LPG cooking technology displaces the use on non-renewable biomass in the baseline which leads to reduced consumption, and this has a positive impact in	No mitigation required
>>			

		reducing deforestation. This will reduce the susceptibility of project areas to natural disasters such as landslides, drought, flooding, or soil erosion.	
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The project activity is an LPG project and does not involve the use of GMOs in its operations.	No mitigation required
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The project entails distribution and installation of metered LPG cooking technology and will not lead to air, water, or land pollution.	No mitigation required
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the	Potentially	The project activity is	The project implementor shall

manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?		promoting use of the smart meter LPG technology. The meters make use of lithium-ion based rechargeable batteries, which have a lifespan of up to 25 years. After the lifespan, the batteries will need to be safely disposed. Under the 5 th schedule of the Environmental Management and Co-Ordination (E-Waste Management) Regulations, 2013 ⁴⁰ , batteries are listed as e-waste and they need to be disposed in line with the provisions of the guidelines.	ensure all the batteries that reach their end of lifespan are collected, recorded, and taken for safe disposal at a NEMA approved recycling site for safe disposal.
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The project involves installation and distribution of the metered LPG cooking	No mitigation required

⁴⁰ [Draft E-waste Regulations-1.pdf \(nema.go.ke\)](#)

>>		technology and does not involve use of pesticides nor fertilizers.	
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The project involves installation and distribution of the metered LPG cooking technology and does not entail harvesting of forests.	No mitigation required
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The project involves installation and distribution of the metered LPG cooking technology and does not have any implication of food quantity or nutritional quality.	No mitigation required
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The project involves installation and distribution of metered LPG cooking technology and does not involve animal husbandry.	No mitigation required
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically	No	The project involves	No mitigation required

affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?		installation and distribution of the metered LPG cooking technology and will not affect or alter any high conservation areas or critical habitats.	
>>			
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?	No	The project involves installation and distribution of the complete metered LPG cooking technology in end-user premises and is not expected to have any impact on endangered species.	No mitigation required
b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?			
>>			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	M Gas Limited
Registration number with relevant authority	108446 (certificate of registration of trademark)
Street/P.O. Box	27780 -00100 NAIROBI
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Country	KENYA
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E-mail	customercare@mgas.ke
Website	https://mgas.ke/
Contact person	
Title	COUNTRY MANAGER
Salutation	MR.
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Middle name	
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Department	MANAGEMENT
Mobile	
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APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to [Design Changes Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.1	7 October 2020	- Hyperlinked section summary to enable quick access to key sections - Improved clarity on Key Project Information - Inclusion criteria table added - Gender sensitive requirements added - Prior consideration (1 yr rule) and Ongoing Financial Need added - Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity - Improved Clarity on SDG contribution/SDG Impact term used throughout - Clarity on Stakeholder Consultation information required - Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption

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